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AXI4-Stream FIFO LogiCORE IP Product Guide (PG080)

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Introduction

Features

IP Facts

Core Overview

Applications

Unsupported Features

Licensing and Ordering

Product Specification

Standards

Performance

Resource Utilization

Port Descriptions

Register Space

Designing with the Core

General Design Guidelines

Clocking

Resets

Protocol Description

Programing Sequence

Design Flow Steps

Customizing and Generating the Core

Constraining the Core

Simulation

Synthesis and Implementation

Verification, Compliance, and Interoperability

Simulation

Hardware Testing

Upgrading

Migrating to the Vivado Design Suite

Upgrading in the Vivado Design Suite

Debugging

Finding Help with AMD Adaptive Computing Solutions

Debug Tools

Additional Resources and Legal Notices

Finding Additional Documentation

Support Resources

References

Revision History

Please Read: Important Legal Notices

AXI4-Stream FIFO LogiCORE IP Product Guide (PG080)

Introduction

The LogiCORE IP AXI4-Stream FIFO core allows memory mapped access to an AXI4-Stream interface. The core can be used to interface to AXI4-Stream IPs, similar to the AXI4-Stream FIFO core, without having to use a full DMA solution.

The principal operation of this core allows the write or read of data packets to or from a device without concern over the AXI4-Stream interface signaling. You can easily manage the AXI4-Stream interfaces as they are transparent.

Features

- 32-bit AXI4-Lite slave interface.
- Configurable data interface type (AXI4 or AXI4-Lite).
- Configurable 32, 64, 128, 256, or 512-bit AXI4-Stream data interface. For AXI4-Lite, the FIFO data width is 32 bits. For AXI4, the FIFO data width matches the AXI4-Stream data width.
- Independently configurable internal TX and RX data FIFOs.
- Full duplex operation.
- Supports AXI Ethernet basic mode.
- Provides interrupts for error and status conditions.
- TX and RX cut-through mode.
- ECC enable/disable option for TX and RX FIFO.

IP Facts

AMD LogiCORE™ IP Facts Table

Core Specifics

AMD LogiCORE™ IP Facts Table

Supported Device Family ¹	AMD Versal™ adaptive SoC AMD UltraScale+™ families AMD UltraScale™ families AMD Zynq™ 7000 SoC 7 series FPGAs
Supported User Interfaces	AXI4, AXI4-Lite, AXI4-Stream
Resources	See Resource Utilization
Provided with Core	
Design Files	VHDL
Example Design	VHDL
Test Bench	VHDL
Constraints File	Not Provided
Simulation Model	Not Provided
Supported S/W Driver ²	Standalone
Tested Design Flows ³	
Design Entry	Vivado Design Suite
Simulation	For support simulators, see the Vivado Design Suite User Guide: Release Notes, Installation, and Licensing (UG973)
Synthesis	Vivado Synthesis
Support	
Release Notes and Known Issues	Master Answer Record: 54447

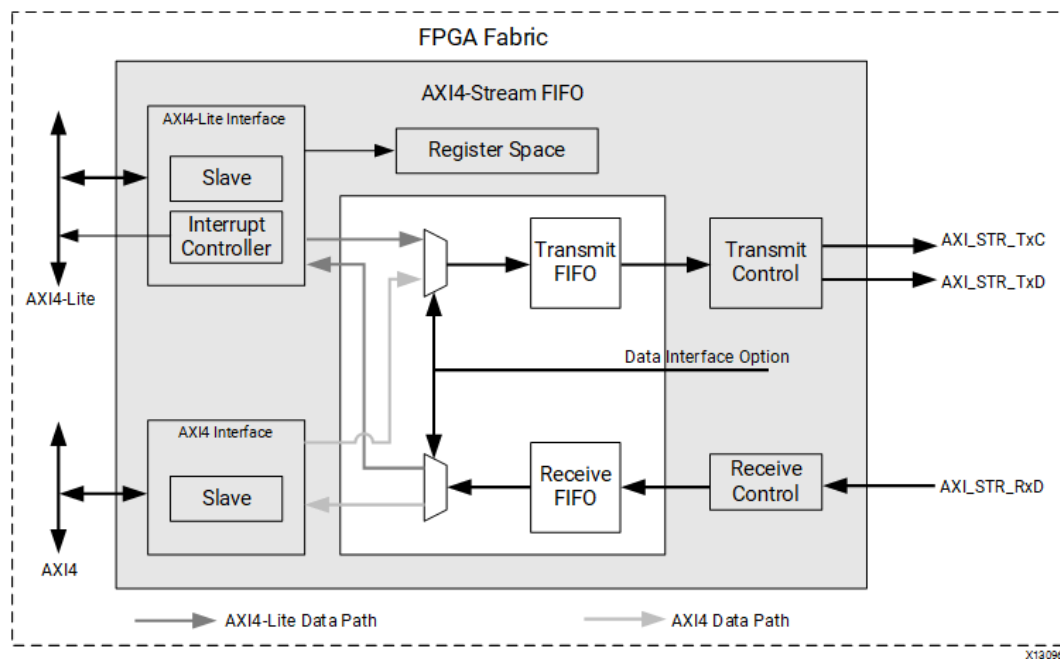
AMD LogiCORE™ IP Facts Table	
All Vivado IP Change Logs	Master Vivado IP Change Logs: 72775
Support web page	
1. For a complete list of supported devices, see the Vivado IP catalog. 2. Linux OS and driver support information is available from the Linux AXI4-Stream FIFO Standalone Driver Page. 3. For the supported versions of third-party tools, see the Vivado Design Suite User Guide: Release Notes, Installation, and Licensing (UG973).	

Core Overview

The following diagram shows the major components of the AXI4-Stream FIFO core:

- AXI Interface block with an AXI4/AXI4-Lite Slave interface
- Interrupt controller
- Registers space
- Receive control module
- Transmit control module
- Receive FIFO for the receive data and length
- Transmit FIFO for the transmit data and the length

Figure: AXI4–Stream FIFO Core Block Diagram



Note: Supported data widths for

AXI4_STR_TxC/AXI4_STR_TxD/AXI4_STR_RxD are 32/64/128/256/512 bits.

The AXI4–Stream FIFO core was designed to provide memory-mapped access to an AXI4–Stream interface connected to other IP (such as the AXI Ethernet core). Systems must be built through the AMD Vivado™ Design Suite to attach the AXI4–Stream FIFO core, AXI Ethernet core, processor, memory, interconnect the buses, clocking, and additional embedded components.

Applications

The AXI4–Stream FIFO core converts AXI4/AXI4–Lite transactions to and from AXI4–Stream transactions, and can be used in Ethernet applications and others that use packet communication.

Unsupported Features

This core does not support asynchronous clock (independent clock) mode.

Licensing and Ordering

This AMD LogiCORE™ IP module is provided at no additional cost with the AMD Vivado™ Design Suite under the terms of the End User License.

Information about this and other LogiCORE IP modules is available at the [Intellectual Property](#) page. For information on pricing and availability of other LogiCORE IP modules and tools, contact your AMD sales representative.

Product Specification

This chapter includes the following sections:

- [Standards](#)
- [Performance](#)
- [Resource Utilization](#)
- [Port Descriptions](#)
- [Register Space](#)

Standards

This core complies with both the AMBA® AXI4–Stream Protocol Specification and the AMBA® AXI4 Protocol Specification.

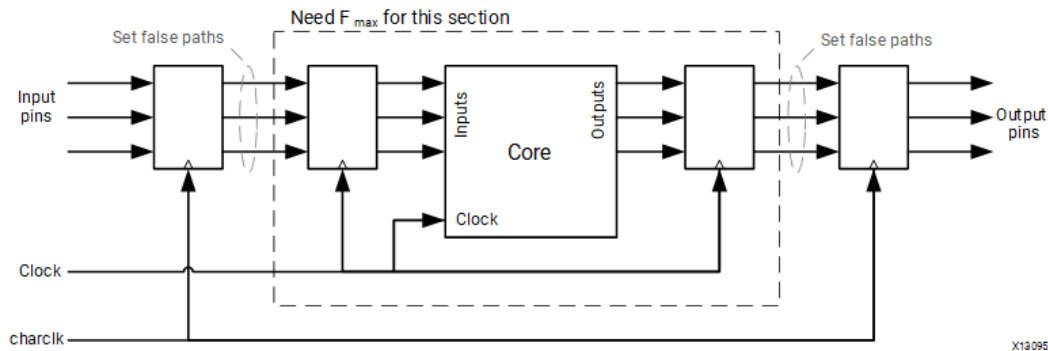
Performance

To measure the performance (F_{MAX}) of the AXI4–Stream FIFO core, it was added as the Device Under Test (DUT) to an AMD Virtex™ 7 FPGA as shown in the following figure.

Because the core is used without other design modules in the FPGA, the utilization and timing numbers reported in this section are estimates only. When this core is combined with other designs in the system, the utilization

of FPGA resources and timing of the design can vary from the results reported here.

Figure: Virtex 7 FPGA with the AXI4–Stream FIFO Core



Maximum Frequencies

When targeting Virtex 7 devices, the AXI4–Stream FIFO core can operate up to 300 MHz with the following configuration:

- Data Interface Option: AXI4
- AXI4 Data Width: 64
- Transmit/Receive FIFO Depth: 4096 locations

The F_{MAX} is influenced by the exact system and provided for guidance. It is not a guaranteed value across all systems.

Latency

Depending on the configuration of the core, the latency between AXI4 to AXI4–Stream varies. For example, if the core is configured for store and forward mode, AXI4–Stream transactions might not start until the entire packet is written into the Transmit FIFO from the AXI4–Lite/AXI4 interface. If the core is configured for TX cut-through/RX cut-through mode, the AXI4–Stream transactions start three clocks after WDATA is accepted from the AXI4–Lite/AXI4 interface.

Throughput

The throughput varies depending on the configuration of the AXI4-Stream FIFO core. For example, if the Data Interface is configured as AXI4-Lite, the throughput is less because the core can accept WDATA once in three clock cycles. If the Data Interface is configured as AXI4, the throughput is three times more than the AXI4-Lite interface. The following table shows the throughput numbers for store-and-forward mode and cut-through mode. Clock frequency is kept constant at 100 MHz to calculate the throughput.

Table: AXI4-Stream FIFO Transmit Throughput

Interface	Frequency (In MHz)	Packet Length (In Bytes)	Maximum Data Throughput (MBytes/sec)	
			Store-and-Forward Mode	Cut-Through Mode
AXI4-Lite	100	8 KB	64.88	77.51
AXI4	100	8 KB	198.49	393.65

Table: AXI4-Stream FIFO Receive Throughput

Interface	Frequency (In MHz)	Packet Length (In Bytes)	Maximum Data Throughput (MBytes/sec)	
			Store-and-Forward Mode	Cut-Through Mode
AXI4-Lite	100	8 KB	66.24	78.99
AXI4	100	8 KB	196.97	370.09

Resource Utilization

For details about resource utilization, visit [Performance and Resource Utilization](#).

Port Descriptions

The AXI4-Stream FIFO core has three AXI4-Stream interfaces: one for transmitting data, one for transmit control, and one for receiving data.

When using the AXI4-Stream FIFO core with the AXI Ethernet core, connect the three AXI4-Stream interfaces listed:

- AXI_STR_TXD – AXI4-Stream Transmit Data
- AXI_STR_TXC – AXI4-Stream Transmit Control
- AXI_STR_RXD – AXI4-Stream Receive Data

The AXI4-Stream Transmit Control Interface supports the transmit protocol of AXI Ethernet cores. The AXI4-Stream Transmit Control Interface is used by the AXI Ethernet core for partial CSUM off-loading of extended VLAN features. The AXI4-Stream FIFO core does not support any advanced features and drives constant values on this interface. The AXI4-Stream FIFO core follows the handshake requirements as defined by the AXI Ethernet Core. For more details, see the AXI 1G/2.5G Ethernet Subsystem Product Guide ([PG138](#)).

The AXI4-Stream FIFO core uses one clock from the AXI4-Lite interface for all clock inputs. When the AXI Ethernet core is used with the AXI4-Stream FIFO core, all the AXI4-Stream input clocks of the AXI Ethernet core must use the same clock.

Table: I/O Signals

Signal Name	I/O	Width	Default	Description
Top-Level System Signal				

Signal Name	I/O	Width	Default	Description
Interrupt	O	1	0	Indicates Interrupt status. This signal is asserted if any of the interrupt status bit is set to 1 and interrupt is enabled for that particular bit. By default, the signal is set to 0.
Global Signals				
s_axi_aclk	I	1	0	Global Interface Clock: All signals on Interface must be synchronous to ACLK.

Signal Name	I/O	Width	Default	Description
s_axi_aresetn	I	1	0	Global reset: This signal is active-Low, ARESETN must be asserted at least for one clock cycle.
AXI4-Lite Write Address Channel Signals				
s_axi_awaddr	I	C_S_AXI_ADDR_WIDTH	0	Write Address: The write address bus gives the address of the first transfer in a write burst transaction. The associated control signals are used to determine the addresses of the remaining transfers in the burst.

Signal Name	I/O	Width	Default	Description
s_axi_awvalid	I	1	0	Write Address Valid: Indicates that valid write address and control information are available: • 1 = Address and control information available. • 0 = Address and control information not available. The address and control information remain stable until the address

Signal Name	I/O	Width	Default	Description
				acknowledge signal, AWREADY, goes High.
s_axi_awready	O	1	0	Write Address Ready: Indicates that the slave is ready to accept an address and associated control signals: • 1 = Slave ready • 0 = Slave not ready
AXI4-Lite Write Data Channel Signals				
s_axi_wdata	I	C_S_AXI_DATA_WIDTH	0	Write Data: The write data bus width is 32-bit only.

Signal Name	I/O	Width	Default	Description
s_axi_wstrb	I	C_S_AXI_DATA_0 WIDTH / 8	0	Write Strobes: Indicates which byte lanes to update in memory. There is one write strobe for each eight bits of the write data bus. Therefore, WSTRB[n] corresponds to WDATA[(8 × n) + 7:(8 × n)]. For example: • S_AXI_WSTRB[0] = 1, WDATA[7:0] is valid. • S_AXI_WSTRB[3] = 0b, WDATA[31:24] is not valid.

Signal Name	I/O	Width	Default	Description
s_axi_wvalid	I	1	0	Write Valid: Indicates that valid write data and strobes are available: • 1 = Write data and strobes available • 0 = Write data and strobes not available

Signal Name	I/O	Width	Default	Description
s_axi_wready	O	1	0	Write Ready: Indicates that the slave can accept the write data: • 1 = Slave ready • 0 = Slave not ready
AXI4-Lite Write Response Channel Signals				
s_axi_bresp	O	2	0	Write Response: Indicates the status of the write transaction. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR.

Signal Name	I/O	Width	Default	Description
s_axi_bvalid	O	1	0	Write Response Valid: Indicates that a valid write response is available: • 1 = Write response available • 0 = Write response not available

Signal Name	I/O	Width	Default	Description
s_axi_bready	I	1	1	Response Ready: Indicates that the master can accept the response information. • 1 = Master ready • 0 = Master not ready
AXI4-Lite Read Address Channel Signals				

Signal Name	I/O	Width	Default	Description
s_axi_araddr	I	C_S_AXI_ADDR_WIDTH	0	Read Address: The read address bus gives the initial address of a read burst transaction. Only the start address of the burst is provided and the control signals that are issued alongside the address detail how the address is calculated for the remaining transfers in the burst.

Signal Name	I/O	Width	Default	Description
s_axi_arvalid	I	1	0	Read Address Valid: When High, indicates that the read address and control information is valid and it remains stable until the address acknowledge signal, ARREADY, is High. • 1 = Address and control information valid • 0 = Address and control information not valid

Signal Name	I/O	Width	Default	Description
s_axi_arready	O	1	0	Read Address Ready: Indicates that the slave is ready to accept an address and associated control signals: • 1 = Slave ready • 0 = Slave not ready
AXI4-Lite Read Data Channel Signals				
s_axi_rdata	O	C_S_AXI_DATA_WIDTH	0	Read Data: The read data bus width is 32-bit only.

Signal Name	I/O	Width	Default	Description
s_axi_rresp	O	2	0	Read Response: Indicates the status of the read transfer. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR.
s_axi_rvalid	O	1	0	Read Valid: Indicates that the required read data is available and the read transfer can complete: • 1 = Read data available • 0 = Read data not available

Signal Name	I/O	Width	Default	Description
s_axi_rready	I	1	0	Read Ready: Indicates that the master can accept the read data and response information: • 1 = Master ready • 0 = Master not ready
AXI4-Stream Transmit Data Channel Signals				
axi_str_txd_aclk	I	1	0	ACLK: Clock for the AXI4-Stream Transmit data interface. Presently not used in the core.

Signal Name	I/O	Width	Default	Description
mm2s_prmry_reset_out_n	O	1	0	Reset: Reset for the AXI4-Stream Transmit data interface.
axi_str_txd_tvalid	O	1	0	TVALID: Indicates that the master is driving a valid transfer. A transfer takes place when both TVALID and TREADY are asserted.
axi_str_txd_tready	I	1	0	TREADY: Indicates that the slave can accept a transfer in the current cycle.

Signal Name	I/O	Width	Default	Description
axi_str_txd_tdata	O	C_S_AXI_DATA_WIDTH Or C_S_AXI4_DATA_WIDTH	0	TDATA: The primary payload that is used to provide the data that is passing across the interface. The width of the data payload is an integer number of bytes. Supported TDATA widths include: 32, 64, 128, 256 or 512 (AXI4 interface only).

Signal Name	I/O	Width	Default	Description
axi_str_txd_tkeep	O	C_S_AXI_DATA_WIDTH/8 Or C_S_AXI4_DATA_WIDTH/8	0	TKEEP: The byte qualifier that indicates whether the content of the associated byte of TDATA is valid. For a 32-bit DATA, bit 0 corresponds to the least significant byte on DATA, and bit 3 corresponds to the most significant byte.
axi_str_txd_tlast	O	1	0	TLAST: Indicates the boundary of a packet.

Signal Name	I/O	Width	Default	Description
axi_str_txd_tdest	O	C_AXIS_TDEST_WIDTH	0	TDEST: Destination AXI4-Stream Identifier and provides routing information for the data stream.
axi_str_txd_tstrb	O	C_S_AXI_DATA_WIDTH/8 Or C_S_AXI4_DATA_WIDTH/8	0	TSTRB: The byte qualifier that indicates whether the content of the associated byte of TDATA is processed as a data byte or a position byte.
axi_str_txd_tid	O	C_AXIS_TID_WIDTH	0	TID: The data stream identifier that indicates different streams of data.

Signal Name	I/O	Width	Default	Description
axi_str_txd_tuser	O	C_AXIS_TUSER_WIDTH	0	TUSER: User-defined sideband information that can be transmitted with the data stream.
AXI4-Stream Transmit Control Channel Signals				
axi_str_txc_aclk	I	1	0	ACLK: Clock for the AXI4-Stream Transmit data interface. Presently not used in the core.
mm2s_cntrl_reset_out_n	O	1	0	Reset: Reset for the AXI4-Stream Transmit data interface.

Signal Name	I/O	Width	Default	Description
axi_str_txc_tvalid	O	1	0	TVALID: Indicates that the master is driving a valid transfer. A transfer takes place when both TVALID and TREADY are asserted
axi_str_txc_tready	I	1	0	TREADY: Indicates that the slave can accept a transfer in the current cycle

Signal Name	I/O	Width	Default	Description
axi_str_txc_tdata	O	C_S_AXI_DATA_0 WIDTH Or C_S_AXI4_DATA_WIDTH	0	TDATA: The primary payload that is used to provide the data that is passing across the interface. The width of the data payload is an integer number of bytes. Supported TDATA widths include 32, 64, 128, 256, or 512 (AXI4 Interface only).

Signal Name	I/O	Width	Default	Description
axi_str_txc_tkeep	O	C_S_AXI_DATA_0 WIDTH/8 Or C_S_AXI4_DATA_WIDTH/8	0	TKEEP: The byte qualifier that indicates whether the content of the associated byte of TDATA is valid. For a 32-bit DATA, bit 0 corresponds to the least significant byte on DATA, and bit 3 corresponds to the most significant byte.
axi_str_txc_tlast	O	1	0	TLAST: Indicates the boundary of a packet.

Signal Name	I/O	Width	Default	Description
axi_str_txc_tstrb	O	C_S_AXI_DATA_WIDTH/8 Or C_S_AXI4_DATA_WIDTH/8	0	TSTRB: The byte qualifier that indicates whether the content of the associated byte of TDATA is processed as a data byte or a position byte.
axi_str_txc_tid	O	C_AXIS_TID_WIDTH	0	TID: The data stream identifier that indicates different streams of data.
axi_str_txc_tdest	O	C_AXIS_TDEST_WIDTH	0	TDEST: Provides routing information for the data stream.

Signal Name	I/O	Width	Default	Description
axi_str_txc_tuser	O	C_AXIS_TUSER_WIDTH	0	TUSER: User-defined sideband information that can be transmitted with the data stream
AXI4-Stream Receive Data Channel Signals				
axi_str_rxd_aclk	I	1	0	ACLK: Clock for the AXI4-Stream Transmit data interface. Presently not used in the core.
s2mm_prry_reset_out_n	O	1	0	Reset: Reset for the AXI4-Stream Transmit data interface.

Signal Name	I/O	Width	Default	Description
axi_str_rxd_tvalid	I	1	0	TVALID: Indicates that the master is driving a valid transfer. A transfer takes place when both TVALID and TREADY are asserted.
axi_str_rxd_tready	O	1	0	TREADY: Indicates that the slave can accept a transfer in the current cycle.

Signal Name	I/O	Width	Default	Description
axi_str_rxd_tdata	I	C_S_AXI_DATA_WIDTH Or C_S_AXI4_DATA_WIDTH	0	TDATA: The primary payload that is used to provide the data that is passing across the interface. The width of the data payload is an integer number of bytes. Supported TDATA widths include 32, 64, 128, 256, or 512 (AXI4 interface only).

Signal Name	I/O	Width	Default	Description
axi_str_rxd_tkeep	I	C_S_AXI_DATA_WIDTH/8 Or C_S_AXI4_DATA_WIDTH/8	0	TKEEP: The byte qualifier that indicates whether the content of the associated byte of TDATA is valid. For a 32-bit DATA, bit 0 corresponds to the least significant byte on DATA, and bit 3 corresponds to the most significant byte.
axi_str_rxd_tlast	I	1	0	TLAST: Indicates the boundary of a packet.

Signal Name	I/O	Width	Default	Description
axi_str_rxd_tdest	I	C_AXIS_TDEST_WIDTH	0	TDEST: Destination AXI4-Stream Identifier and provides routing information for the data stream.
axi_str_rxd_tstrb ¹	O	C_S_AXI_DATA_WIDTH/8 Or C_S_AXI4_DATA_WIDTH/8	0	TSTRB: The byte qualifier that indicates whether the content of the associated byte of TDATA is processed as a data byte or a position byte.
axi_str_rxd_tid ¹	O	C_AXIS_TID_WIDTH	0	TID: The data stream identifier that indicates different streams of data.

Signal Name	I/O	Width	Default	Description
axi_str_rxd_ tuser ¹	O	C_AXIS_TUSER_ WIDTH	0	TUSER: User-defined sideband information that can be transmitted with the data stream.
AXI4 Write Address Channel Signals				
s_axi4_awid	I	C_S_AXI_ID_ WIDTH	0	Write Address ID: Identification tag for the write address group of signals.

Signal Name	I/O	Width	Default	Description
s_axi4_awaddr	I	C_S_AXI_ADDR_WIDTH	0	Write Address: The write address bus gives the address of the first transfer in a write burst transaction. The associated control signals are used to determine the addresses of the remaining transfers in the burst.

Signal Name	I/O	Width	Default	Description
s_axi4_awlen	I	8	0	Burst Length: The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
s_axi4_awsize	I	3	0	Burst Size: Indicates the size of each transfer in the burst.

Signal Name	I/O	Width	Default	Description
s_axi4_awburst	I	2	0	Burst Type: The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated. This core supports incremental burst type only. Core behavior is not guaranteed if unsupported burst type is set.

Signal Name	I/O	Width	Default	Description
s_axi4_awlock	I	1	0	Lock Type: This signal provides additional information about the atomic characteristics of the transfer. Presently this signal is not used in the core.
s_axi4_awcache	I	4	0	Cache Type: Indicates the bufferable, cacheable, writethrough, write-back, and allocate attributes of the transaction. Presently this signal is not used in the core.

Signal Name	I/O	Width	Default	Description
s_axi4_awprot	I	3	0	Protection Type: Indicates the normal, privileged, or secure protection level of the transaction and whether the transaction is a data access or an instruction access. Presently this signal is not used in the core.

Signal Name	I/O	Width	Default	Description
s_axi4_awvalid	I	1	0	Write Address Valid: Indicates that valid write address and control information are available: • 1 = Address and control information available • 0 = Address and control information not available The address and control information remain stable until the address

Signal Name	I/O	Width	Default	Description
				acknowledge signal, AWREADY, goes High.
s_axi4_awready	O	1	0	Write Address Ready: Indicates that the slave is ready to accept an address and associated control signals: • 1 = Slave ready • 0 = Slave not ready
AXI4 Write Data Channel Signals				
s_axi4_wdata	I	C_S_AXI4_DATA_WIDTH	0	Write Data: The write data bus can be 32 or 64 bits wide.

Signal Name	I/O	Width	Default	Description
s_axi4_wstrb	I	C_S_AXI4_DATA_WIDTH/8	0	Write Strobes: Indicates which byte lanes to update in memory. There is one write strobe for each eight bits of the write data bus. Therefore, WSTRB[n] corresponds to WDATA[(8 × n) + 7:(8 × n)]. For a 64-bit DATA, bit 0 corresponds to the least significant byte on DATA, and bit 7 corresponds to the most significant byte. For example:

Signal Name	I/O	Width	Default	Description
				• STROBE [0] = 1b, DATA[7:0] is valid • STROBE [7] = 0b, DATA[63:56] is not valid
s_axi4_wlast	I	1	0	Write Last: Indicates the last transfer in a write burst.

Signal Name	I/O	Width	Default	Description
s_axi4_wvalid	I	1	0	Write Valid: Indicates that valid write data and strobes are available: • 1 = Write data and strobes available • 0 = Write data and strobes not available

Signal Name	I/O	Width	Default	Description
s_axi4_wready	O	1	0	Write Ready: Indicates that the slave can accept the write data: • 1 = Slave ready • 0 = Slave not ready
AXI4 Write Response Channel Signals				
s_axi4_bid	O	C_S_AXI_ID_WIDTH	0	Response ID: The identification tag of the write response. The BID value must match the AWID value of the write transaction to which the slave is responding.

Signal Name	I/O	Width	Default	Description
s_axi4_bresp	O	2	0	Write Response: Indicates the status of the write transaction. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR.
s_axi4_bvalid	O	1	0	Write Response Valid: Indicates that a valid write response is available: • 1 = Write response available. • 0 = Write response not available.

Signal Name	I/O	Width	Default	Description
s_axi4_bready	I	1	1	Response Ready: Indicates that the master can accept the response information. • 1 = Master ready. • 0 = Master not ready.
AXI4 Read Address Channel Signals				

Signal Name	I/O	Width	Default	Description
s_axi4_arid	I	C_S_AXI_ID_ WIDTH	0	Read Address ID: This signal is the identification tag for the read address group of signals. ARID is always set to zero; all configured channels access to a single address MAP region in Memory mapped interconnect.

Signal Name	I/O	Width	Default	Description
s_axi4_araddr	I	C_S_AXI_ADDR_WIDTH	0	Read Address: The read address bus gives the initial address of a read burst transaction. Only the start address of the burst is provided and the control signals that are issued alongside the address detail how the address is calculated for the remaining transfers in the burst.

Signal Name	I/O	Width	Default	Description
s_axi4_arlen	I	8	0	Burst Length: The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
s_axi4_arsize	I	3	0	Burst Size: This signal indicates the size of each transfer in the burst. Burst Size is always set based on configured data width of the interface.

Signal Name	I/O	Width	Default	Description
s_axi4_arburst	I	2	0	Burst Type: The burst type, coupled with the size information, details how the address for each transfer within the burst is calculated. Burst Type is always set to Incremental. Core behavior is not guaranteed if unsupported burst type is set.

Signal Name	I/O	Width	Default	Description
s_axi4_arlock	I	1	0	Lock Type: This signal provides additional information about the atomic characteristics of the transfer. Presently this signal is not used in the core.
s_axi4_arcache	I	4	0	Cache Type: This signal provides additional information about the cacheable characteristics of the transfer. Presently this signal is not used in the core.

Signal Name	I/O	Width	Default	Description
s_axi4_arprot	I	3	0	Protection Type: This signal provides protection unit information for the transaction. Presently this signal is not used in the core.

Signal Name	I/O	Width	Default	Description
s_axi4_arvalid	I	1	0	Read Address Valid: When High, indicates that the read address and control information is valid and it remains stable until the address acknowledge signal, ARREADY, is High. • 1 = Address and control information valid • 0 = Address and control information not valid

Signal Name	I/O	Width	Default	Description
s_axi4_arready	O	1	0	Read Address Ready: Indicates that the slave is ready to accept an address and associated control signals: • 1 = Slave ready • 0 = Slave not ready
AXI4 Read Data Channel Signals				

Signal Name	I/O	Width	Default	Description
s_axi4_rid	O	C_S_AXI_ID_ WIDTH	0	Read ID Tag: ID tag of the read data group of signals. The RID value is generated by the slave and must match the ARID value of the read transaction to which it is responding.
s_axi4_rdata	O	C_S_AXI4_DATA_WIDTH	0	Read Data: The read data bus can be 32 or 64 bits wide.

Signal Name	I/O	Width	Default	Description
s_axi4_rresp	O	2	0	Read Response: Indicates the status of the read transfer. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR.
s_axi4_rlast	O	1	0	Read Last: Indicates the last transfer in a read burst.

Signal Name	I/O	Width	Default	Description
s_axi4_rvalid	O	1	0	Read Valid: Indicates that the required read data is available and the read transfer can complete: • 1 = Read data available • 0 = Read data not available

Signal Name	I/O	Width	Default	Description
s_axi4_rready	I	1	0	Read Ready: Indicates that the master can accept the read data and response information: • 1 = Master ready • 0 = Master not ready
1. This port is currently not used.				

Register Space

The AXI4-Stream FIFO core contains the registers listed in the following table.

Table: Register Names and Descriptions

Register Name	AXI Address	Access
Interrupt Status Register (ISR)	C_BASEADDR + 0x0	Read/Clear on Write ¹
Interrupt Enable Register (IER)	C_BASEADDR + 0x4	Read/Write

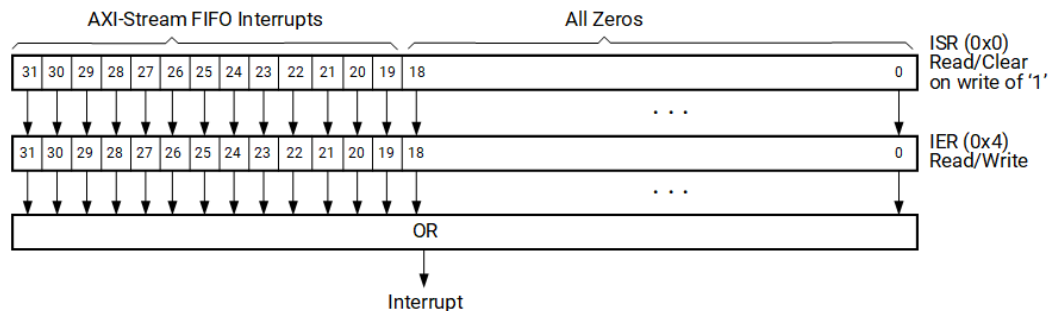
Register Name	AXI Address	Access
Transmit Data FIFO Reset (TDFR)	C_BASEADDR + 0x8	Write ²
Transmit Data FIFO Vacancy (TDFV)	C_BASEADDR + 0xC	Read
Transmit Data FIFO 32-bit Wide Data Write Port (TDFD)	C_BASEADDR + 0x10 or C_AXI4_BASEADDR + 0x0000	Write ³
Transmit Length Register (TLR)	C_BASEADDR + 0x14	Write
Receive Data FIFO reset (RDFR)	C_BASEADDR + 0x18	Write ²
Receive Data FIFO Occupancy (RDFO)	C_BASEADDR + 0x1C	Read
Receive Data FIFO 32-bit Wide Data Read Port (RDFD)	C_BASEADDR + 0x20 or C_AXI4_BASEADDR + 0x1000	Read ³
Receive Length Register (RLR)	C_BASEADDR + 0x24	Read
AXI4-Stream Reset (SRR)	C_BASEADDR + 0x28	Write ²
Transmit Destination Register (TDR)	C_BASEADDR + 0x2C	Write
Receive Destination Register (RDR)	C_BASEADDR + 0x30	Read

Register Name	AXI Address	Access
Transmit ID Register ⁴	C_BASEADDR + x34	Write
Transmit USER Register ⁴	C_BASEADDR + x38	Write
Receive ID Register ⁴	C_BASEADDR + x3C	Read
Receive USER Register ⁴	C_BASEADDR + x40	Read
Transmit FIFO ECC Configuration Register	C_BASEADDR + 0x44	Read Write
Transmit FIFO ECC Error Counter Register	C_BASEADDR + 0x48	Read Only
Receive FIFO ECC Configuration Register	C_BASEADDR + 0x4C	Read Write
Receive FIFO ECC Error Counter Register	C_BASEADDR + 0x50	Read Only
Reserved	C_BASEADDR + 0x54 to C_BASEADDR + 0x7C	N/A ⁵

Register Name	AXI Address	Access
1. The latched interruptible condition is cleared by writing a 1 to that bit location. Writing a 1 to a bit location that is 0 has no effect. Likewise, writing a 0 to a bit location that is 1 has no effect. Multiple bits can be cleared in a single write. 2. Reset if written with 0xA5. 3. C_AXI4_BASEADDR should be used only if the Data Interface option is AXI4. 4. Not currently supported. 5. If read, these registers returns 0x0. Writing these registers has no effect. 6. C_BASEADDR is defined by the interconnect when using IP integrator. When you implement a standalone core (for example, selecting from the IP catalog in Vivado), only the address signals <code>s_axi_awaddr(5:2)</code> and <code>s_axi_araddr(5:2)</code> are decoded. This results in repeating the address map of the registers every 64 hex locations. In this case, <code>C_BASEADDR</code> can be considered to be 0.		

Interrupt Interface

The interrupt signals generated by the AXI4–Stream FIFO core are managed by the ISR and IER registers. The ISR is combined with the IER register to define the interrupt interface of the AXI4–Stream FIFO core. An overview diagram of the interrupt control structure is shown in the following figure.

Figure: Interrupt Control Structure

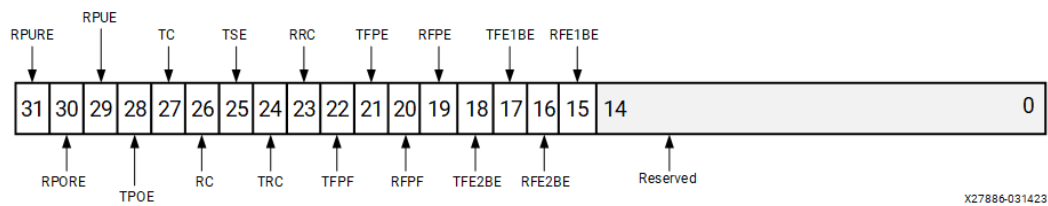
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Interrupt Status Register (ISR)

The Interrupt Status Register is shown in the following figure. The Interrupt Status register uses one bit to represent each internal interruptible condition.

After an interruptible condition occurs, it is captured in this register (represented as the corresponding bit being set to 1) even if the condition changes. The latched interruptible condition is cleared by writing a 1 to its bit location. Writing a 1 to a bit location that is 0 has no effect. Likewise, writing a 0 to a bit location that is 1 has no effect. Multiple bits can be cleared in a single write.

For any bit set in the Interrupt Status Register, a corresponding bit must also be set in the Interrupt Enable Register for the Interrupt signal to be driven active-High out of the AXI4-Stream FIFO core.

Figure: Interrupt Status Register (Offset 0x0)

The Interrupt Status Register bit definitions are detailed in the following table.

Table: Interrupt Status Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
14:0	Reserved	Read	0x0	Reserved: These bits are reserved for future definition and they always return all zeros.
15	RFE1BE	Read/Clear on Write of 1	0	Receive FIFO ECC 1 bit error: This interrupt is generated when a 1 bit error is detected on RX FIFO.

Bit(s)	Name	Core Access	Reset Value	Description
16	RFE2BE	Read/Clear on Write of 1	0	Receive FIFO ECC 2 bit error: This interrupt is generated when a 2 bit error is detected on RX FIFO.
17	TFE1BE	Read/Clear on Write of 1	0	Transmit FIFO ECC 1 bit error: This interrupt is generated when a 1 bit error is detected on TX FIFO.
18	TFE2BE	Read/Clear on Write 1	0	Transmit FIFO ECC 2 bit error: This interrupt is generated when a 2 bit error is detected on TX FIFO.

Bit(s)	Name	Core Access	Reset Value	Description
19	RFPE	Read/Clear on Write of 1	0	Receive FIFO Programmable Empty: Generated when the difference between the read and write pointers of the receive FIFO reaches the programmable EMPTY threshold value when the FIFO is being emptied. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
20	RFPF	Read/Clear on Write of 1	1	Receive FIFO Programmable Full: This interrupt is generated when the difference between the read and write pointers of the receive FIFO reaches the programmable FULL threshold value. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
21	TFPE	Read/Clear on Write of 1	0	Transmit FIFO Programmable Empty: This interrupt is generated when the difference between the read and write pointers of the transmit FIFO reaches the programmable EMPTY threshold value when the FIFO is being emptied. 0 = No interrupt pending 1 = Interrupt pending For lower values of programmable threshold,

Bit(s)	Name	Core Access	Reset Value	Description
				this flag can toggle during the initial writes because of First Word Fall Through (FWFT) behavior of FIFO. This flag can toggle even though there are no external reads.

Bit(s)	Name	Core Access	Reset Value	Description
22	TFPF	Read/Clear on Write of 1	1	Transmit FIFO Programmable Full: This interrupt is generated when the difference between the read and write pointers of the transmit FIFO reaches the programmable FULL threshold value. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
23	RRC	Read/Clear on Write of 1	1	Receive Reset Complete: This interrupt indicates that a reset of the receive logic has completed. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
24	TRC	Read/Clear on Write of 1	1	Transmit Reset Complete: This interrupt indicates that a reset of the transmit logic has completed. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
25	TSE	Read/Clear on Write of 1	0	Transmit Size Error: This interrupt is generated if the number of words (including partial words in the count) written to the transmit data FIFO does not match the value written to the transmit length register (bytes) divided by 4/8 and rounded up to the higher integer value for trailing byte fractions. Interrupts occur only for mismatch of word count

Bit(s)	Name	Core Access	Reset Value	Description
				(including partial words). Interrupts do not occur due to mismatch of byte count. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
26	RC	Read/Clear on Write of 1	0	Receive Complete: Indicates that at least one successful receive has completed and that the receive packet data and packet data length is available. This signal is not set for unsuccessful receives. This interrupt can represent more than one packet received, so it is important to check the receive data FIFO occupancy value to determine if additional receive

Bit(s)	Name	Core Access	Reset Value	Description
				packets are ready to be processed. 0 = No interrupt pending 1 = Interrupt pending
27	TC	Read/Clear on Write of 1	0	Transmit Complete: Indicates that at least one transmit has completed. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
28	TPOE	Read/Clear on Write of 1	0	Transmit Packet Overrun Error: This interrupt is generated if an attempt is made to write to the transmit data FIFO when it is full. A reset of the transmit logic is required to recover. 0 = No interrupt pending 1 = Interrupt pending

Bit(s)	Name	Core Access	Reset Value	Description
29	RPUE	Read/Clear on Write of 1	0	Receive Packet Underrun Error: This interrupt occurs when an attempt is made to read the receive FIFO when it is empty. The data read is not valid. A reset of the receive logic is required to recover. 0 = No interrupt pending 1 = Interrupt pending

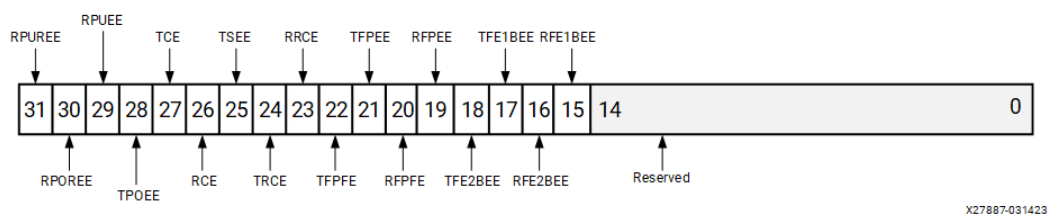
Bit(s)	Name	Core Access	Reset Value	Description
30	RPORE	Read/Clear on Write of 1	0	Receive Packet Overrun Read Error: This interrupt occurs when more words are read from the receive data FIFO than are in the packet being processed. Even though the FIFO is not empty, the read has gone beyond the current packet and removed the data from the next packet. A reset of the receive logic is required to recover. 0 = No interrupt

Bit(s)	Name	Core Access	Reset Value	Description
				pending 1 = Interrupt pending
31	RPURE	Read/Clear on Write of 1	0	Receive Packet Underrun Read Error: This interrupt occurs when an attempt is made to read the receive length register when it is empty. The data read is not valid. A reset of the receive logic is required to recover. 0 = No interrupt pending 1 = Interrupt pending

Interrupt Enable Register (IER)

The Interrupt Enable Register shown in determines which interrupt sources in the Interrupt Status Register are allowed to generate interrupts. Setting to “1” in a bit location enables the related interrupt from being propagated, while a value of “0” disables it.

Figure: Interrupt Enable Register (Offset 0x4)



Transmit Data FIFO Reset Register (TDFR)

The Transmit Data FIFO Reset Register shown below is not an actual register, but is instead a write-only address, which when written with a specific value, generates a reset for the Transmit Data FIFO. This reset does not occur until transmit activity on the TX AXI4-Stream has completed. The reset can occur only during inactive times on the TX AXI4-Stream and it affects only the transmit circuitry in this core, thereby preventing the core on the other end of the AXI4-Stream from receiving a partial packet which could potentially cause a failure condition in the latter core. The reset is applied only during the inactive times on the TX AXI4-Stream. Writing a TDFR register with other than A5 value disables the reset.

Because of this mode of operation, it is possible that if the AXI4-Stream becomes unresponsive during an AXI4-Stream transaction, a reset can never occur. For example, this might occur while waiting for the destination

ready to go active in the middle of a transfer. In such cases it is necessary to use both the AXI4–Stream Reset and the S_AXI_ARESETN reset.

Note: To apply the AXI4–Stream Reset, you must write the SRR register and then assert S_AXI_ARESETN.

Figure: Transmit Data FIFO Reset Register (Offset 0x8)

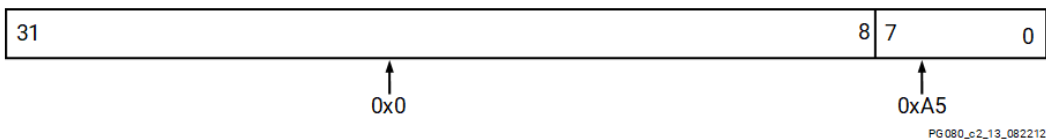


Table: Transmit Data FIFO Reset Register Bit Definitions

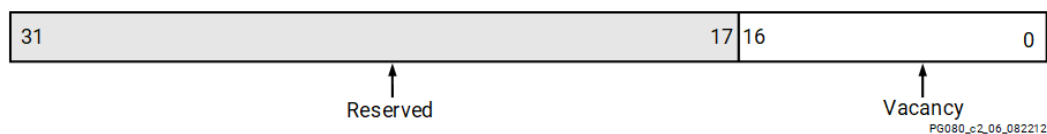
Bit(s)	Name	Core Access	Reset Value	Description
31:0	Reset Key	Write	N/A	Reset Write Value . "0x000000A5" – Generate a reset. Others – No effect.

Transmit Data FIFO Vacancy Register (TDFV)

The Transmit Data FIFO Vacancy Register shown in the figure below is a read-only register that gives the vacancy status of the Transmit Data FIFO. This is an unsigned value and reflects the current snapshot of the number

of locations free for data storage in the Transmit Data FIFO. The value reflected (N) in this register tells you that you can perform N writes to Transmit FIFO. The value of this register after reset is C_TX_FIFO_DEPTH-4. The register does not decrement for every Transmit Data FIFO Data Write Port (TDFD) write. It decrements by two for every two write locations.

Figure: Transmit Data FIFO Vacancy Register (Offset 0xC)



Note: Data written to TDFD register will be written to FIFO with 6 axi clock cycles Transaction Delay . TDFV register will be updated when the written data is reflected in the Transmit Data FIFO.

Transmit Data FIFO Data Write Port (TDFD)

The Transmit Data FIFO Data Write Port shown in is an N-bit wide address location for writing data into the Transmit Data FIFO. N is equal to C_S_AXI_DATA_WIDTH.

Figure: Transmit Data FIFO Data Write Port (Offset 0x10)

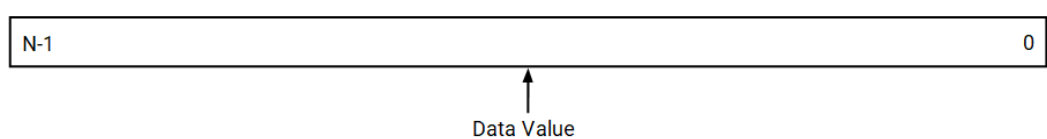


Table: Transmit Data FIFO Data Write Port Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
(N-1) – 0	Write Data Value	Write	N/A	Transmit Data FIFO Write Value . N is equal to C_S_AXI_DATA_WIDTH.

Receive Data FIFO Reset Register (RDFR)

The Receive Data FIFO Reset Register shown in is not an actual register but, rather a write-only address, which when written with a specific value, generates a reset for the Receive Data FIFO.

This reset does not occur until the receive activity on the RX AXI4-Stream has completed. Only during inactive times on the RX AXI4-Stream can a reset occur. It affects only the receive circuitry in this core. This prevents the core on the other end of the AXI4-Stream from transmitting a partial packet which can cause failure condition in that core. The reset is applied only during the inactive times on the RX AXI4-Stream. Writing a RDFR register with other than A5 value disables the reset.

Because of this mode of operation, it is possible that if the AXI4-Stream interface becomes unresponsive during an AXI4-Stream transaction, that the reset does not occur. For example, if a packet is received over the AXI4-Stream that exceeds the FIFO size of this core, the core destination is ready to become inactive in the middle of a transfer. In this case, an S_AXI_ARESETN reset is needed.

Figure: Receive Data FIFO Reset Register (Offset 0x18)

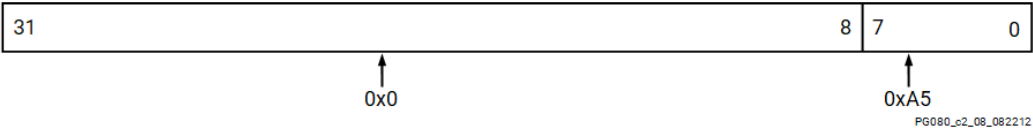


Table: Receive Data FIFO Reset Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
31:0	Reset Key	Write	N/A	Reset Write Value : "0x000000 A5" – Generate a reset. Others – No effect.

Receive Data FIFO Occupancy Register (RDFO)

The Receive Data FIFO Occupancy Register shown in is a read-only register that gives the occupancy status of the Receive Data FIFO.

Figure: Receive Data FIFO Occupancy Register (Offset 0x1C)

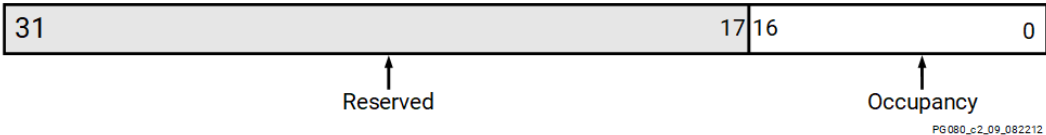


Table: Receive Data FIFO Occupancy Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
--------	------	-------------	-------------	-------------

Bit(s)	Name	Core Access	Reset Value	Description
16:0	Occupancy	Read	0x0	Receive Data FIFO Occupancy: This is the unsigned value reflecting a current snapshot of the number of locations in use for data storage in the receive Data FIFO memory core in the most recent transaction. This value is only updated after a packet is successfully received, and therefore can be used to determine if a receive packet is ready to be processed

Bit(s)	Name	Core Access	Reset Value	Description
				when a non-0 value is read. If the number of packets received is one, then this register returns the value of the locations occupied. After the FIFO is read, any subsequent read to this register returns the value of 0. If more than one packet is received, a read to this register returns the number of locations occupied by the latest received packet.

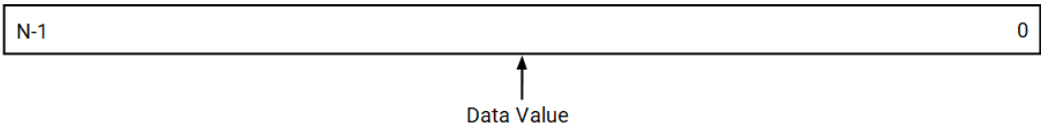
Bit(s)	Name	Core Access	Reset Value	Description
31:17	Reserved	Read	0x0	Reserved: These bits are reserved for future definition and always return all zeros.

Receive Data FIFO Data Read Port (RDFD)

The Receive Data FIFO Data Read Port shown in the following figure is a N-bit wide address location for reading data from the Receive Data FIFO. N is equal to C_S_AXI_DATA_WIDTH.

Important: The value of this register is not guaranteed if this register is read when Receive Packet Underrun Error (RPUE) interrupt bit in Interrupt Status Register (ISR) is set.

Figure: Receive Data FIFO Data Read Port (Offset 0x20)



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Transmit Length Register (TLR)

The Transmit Length Register shown in [Store-and-Forward Mode](#) and [Cut-Through Mode](#). This register is used to store packet length values (the

number of bytes in the packet) corresponding to valid packets ready for transmit. The data for the packet is stored in the transmit Data FIFO. The data is written to the AXI4-Stream FIFO core over the AXI4 interface, typically by a processor or DMA core such the Central DMA (CDMA). When presenting a transmit packet to the AXI4-Stream FIFO core, write the packet data to the Transmit Data FIFO first, then write the length of the packet into the TLR.

It is not valid to write data for multiple packets to the transmit data FIFO before writing the packet length values.

Store-and-Forward Mode

In this mode, packet transmission on the TX AXI4-Stream interface does not start until the TLR is written with a valid packet length value. The width of the TLR is wide enough to support packets up to 8 MBytes-4 in length. The smallest packet that can be transmitted is 1 byte. The maximum packet that can be transmitted is limited by the size of the FIFO, which is $(C_TX_FIFO_DEPTH-4) * (data\ interface\ width / 8)$ bytes.

Figure: Transmit Length Register: Store-and-Forward Mode (Offset 0x14)

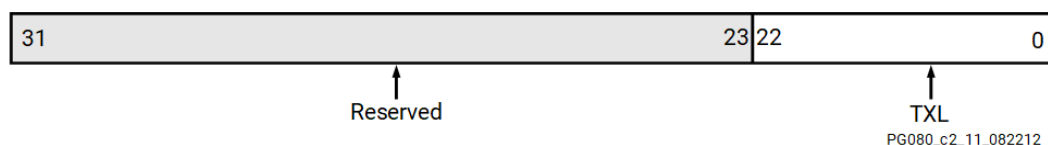


Table: Transmit Length Register (Store-and-Forward Mode) Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
--------	------	-------------	-------------	-------------

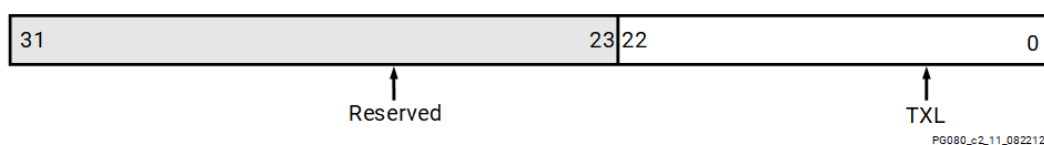
Bit(s)	Name	Core Access	Reset Value	Description
(N-1) – 0	Read Data Value	Read	N/A	Receive Data FIFO Read Value . N is equal to C_S_AXI_DATA_WIDTH.

Cut-Through Mode

In this mode, packet transmission starts on the AXI4-Stream interface when the transmit FIFO is not empty. However, the last beat of the packet is transmitted only when the TLR is written with a valid packet length value.

The width of the TLR is wide enough to support packets up to 8 MBytes-4 (8388604) in length. The smallest packet that can be transmitted is 1 byte. The maximum packet that can be transmitted is 8 MBytes-4 and is independent of the TX FIFO depth selected.

Figure: Transmit Length Register: Cut-Through Mode (Offset 0x14)



Bit(s)	Name	Core Access	Reset Value	Description
22:0	TXL	Write	0x0	Transmit Length : The number of bytes of the corresponding transmit packet stored in the transmit data FIFO.
31:23	Reserved	N/A	0x0	Reserved : These bits are reserved for future definition and always return all zeros.

Receive Length Register (RLR)

The receive length register are shown in [Store-and-Forward Mode](#) and [Cut-Through Mode](#). This register is used to retrieve packet length values (the number of bytes in the packet) corresponding to valid packets received. The data for the packet is stored in the Receive Data FIFO.

The RLR should only be read when a receive packet is available for processing (the receive occupancy is not zero). After the RLR is read, the receive packet data should be read from the receive data FIFO before the RLR is read again.

Note: RDFO should be read before reading RLR. Reading RLR first results in the RDFO being reset to zero.

Store-and-Forward Mode

In this mode, the length is written by the AXI4-Stream FIFO core when the complete packet is received across the RX AXI4-Stream interface. The width of the RLR is wide enough to support packets up to 8 MBytes-4 (8388604) in length. The smallest packet that can be received is 1 byte. The maximum packet that can be received is limited by the size of the FIFO, which is $(C_RX_FIFO_DEPTH-4) \times (\text{data interface width}/8)$ bytes.

Figure: Receive Length Register: Store-and-Forward Mode (Offset 0x24)

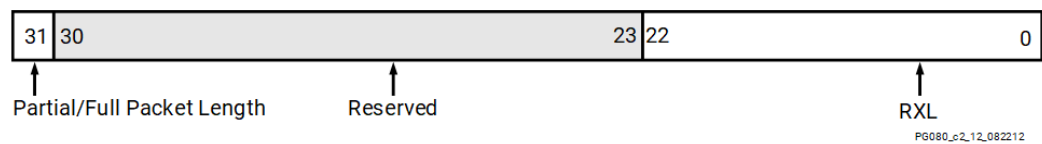


Table: Receive Length Register (Store-and-Forward Mode) Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
22:0	RXL	Read	0x0	Receive Length : The number of bytes of the corresponding receive data stored in the receive data FIFO.

Bit(s)	Name	Core Access	Reset Value	Description
31:23	Reserved	Read	0x0	Reserved : These bits are reserved for future definition and always return all zeros.

Cut-Through Mode

In this mode, the length is written by the AXI4-Stream FIFO core when the RX FIFO is not empty. Bit 31 of the register is used to indicate whether the length value given in the remaining 30 bits is for a partial or full packet. When bit 31 is 1, the length indicates the amount of partial packet data that can be read. After the last beat of the packet received on the AXI4-Stream side, bit 31 becomes 0 and remaining bits show the complete packet length.

The width of the RLR is wide enough to support packets up to 8 MBytes-4 (8388604) in length. The smallest packet that can be received is 1 byte. The maximum packet that can be received is 8 MBytes-4 and is independent of the RX FIFO depth selected in AMD Vivado™ IDE.

Figure: Receive Length Register: Cut-Through Mode (Offset 0x24)

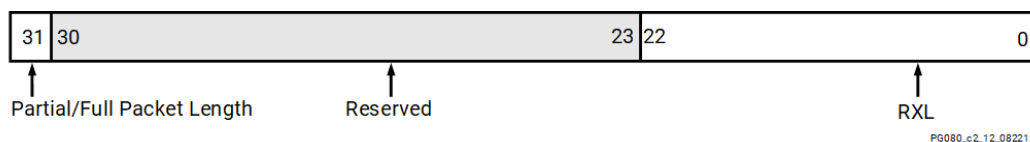


Table: Receive Length Register: (Cut-Through Mode) Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
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Bit(s)	Name	Core Access	Reset Value	Description
22:0	RXL	Read	0x0	Receive Length : The number of bytes of the corresponding receive data stored in the receive data FIFO.
30:23	Reserved	Read	0x0	Reserved : These bits are reserved for future definition and always return all zeros.

Bit(s)	Name	Core Access	Reset Value	Description
31	Partial/Full Packet Length	Read	0x0	Partial/Full Packet Length Indicator: When bit 31 is 1, the RXL indicates the amount of partial packet data that can be read. After the last beat of the packet received on the AXI4-Stream side, bit 31 becomes 0 and the RXL shows the complete packet length.

AXI4-Stream Reset Register (SRR)

The AXI4-Stream Register shown in the following figure is not an actual register. It is a write-only address, which when written with a specific value, generates an immediate reset for the entire core and driving a reset on the external outputs, `s2mm_prmry_reset_out_n`, `mm2s_prmry_reset_out_n`, and `mm2s_cntrl_reset_out_n`, which can be used to reset the core on the other end of the AXI4-Stream.

Figure: AXI4–Stream Reset Register (Offset 0x28)

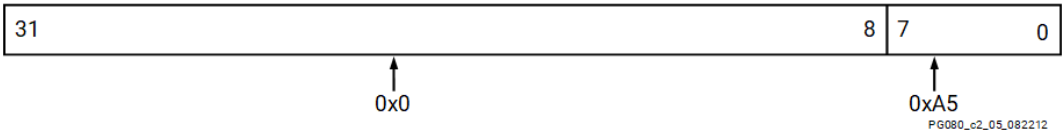


Table: AXI4–Stream Reset Register (Offset 0x28)

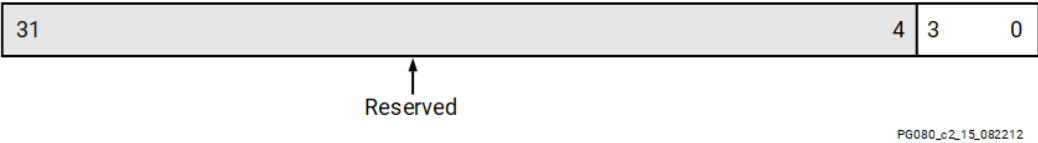
Bit(s)	Name	Core Access	Reset Value	Description
31:0	Reset Key	Write	N/A	Reset Write Value : "0x000000A5" – Generate a reset. Others – No effect.

Transmit Destination Register (TDR)

The Transmit Destination Register shown in stores the destination address corresponding to the packet to be transmitted. When presenting a transmit packet to the AXI4–Stream FIFO core, write the destination address into TDR first, write the packet data to the Transmit Data FIFO next, and then write the length of the packet into the TLR.

The destination address must be written to the TDR before the packet data is written to the transmit data FIFO. Writing data for multiple packets to the transmit data FIFO before writing the destination address values is not a valid sequence.

Figure: Transmit Destination Register (Offset 0x2C)



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Table: Transmit Destination Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
--------	------	-------------	-------------	-------------

Bit(s)	Name	Core Access	Reset Value	Description
3:0	TDEST	Write	0x0	Transmit Destination: The destination address of the transmit packet stored in the transmit data FIFO. TDEST is optional when generating the core and that TDEST can be specified to be 1 to 4 bits when it is implemented (see TDEST and corresponding “Width” in .
31:4	Reserved	N/A	0x0	Reserved : These bits are reserved for future definition and always return all zeros.

Receive Destination Register (RDR)

The Receive Destination Register shown in retrieves the destination address corresponding to the valid packet received.

The RDR should only be read when a receive packet is available for processing (the receive occupancy is not zero). After the RDR is read, the receive packet data should be read from the receive data FIFO before the RDR is read again. The RDR values are stored in the receive data FIFO by the AXI4-Stream FIFO core with the data of each packet. The RDR value for the subsequent packet to be processed is moved to the RDR when the previous RDR value has been read.

Figure: Receive Destination Register (Offset 0x30)

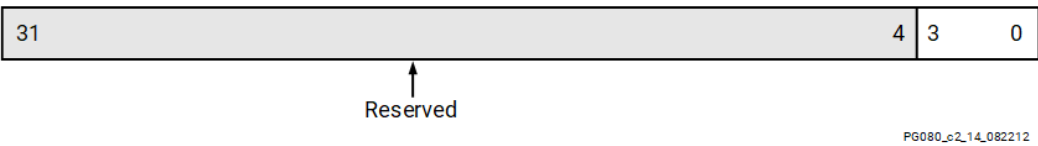


Table: Receive Destination Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
--------	------	-------------	-------------	-------------

Bit(s)	Name	Core Access	Reset Value	Description
3:0	RDEST	Read	0x0	Receive Destination : The destination address of the receive packet stored in the receive data FIFO. TDEST is optional when generating the core and that TDEST can be specified to be 1 to 4 bits when it is implemented (see TDEST and corresponding "Width" in .
31:4	Reserved	N/A	0x0	Reserved : These bits are reserved for future definition and always return all zeros.

Transmit FIFO ECC Configuration Register

The Transmit FIFO ECC Configuration Register shown in the following table is not an actual register, but is instead a write-only address. Bit 0 is for resetting the Transmit FIFO ECC Error counters, other bits are reserved for internal and future use.

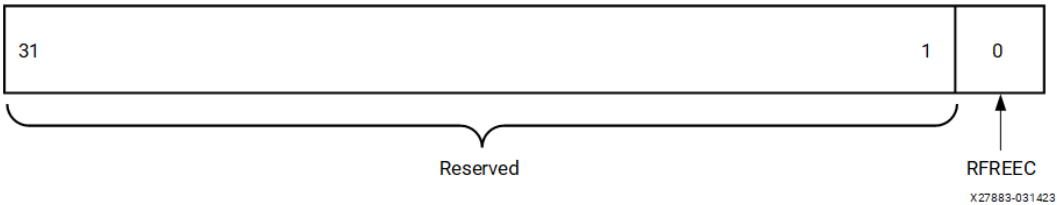
ECC Error counters Reset is applied as long as the bit 0 is '1'.

Table: AXI4-Stream Transmit FIFO ECC Configuration Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
31:1	N/A	Read	0	These bits are reserved for future definition and always return all zeros.

Bit(s)	Name	Core Access	Reset Value	Description
0	TFREEC	Read/Write	0	Transmit FIFO Reset ECC Error Counters: Reset ECC error counters for Transmit FIFO. Write 1 to clear the Transmit FIFO ECC Error counters. Counters are cleared as long as this bit is '1'.

Figure: AXI4–Stream Transmit FIFO ECC Configuration Register (Offset 0x44)



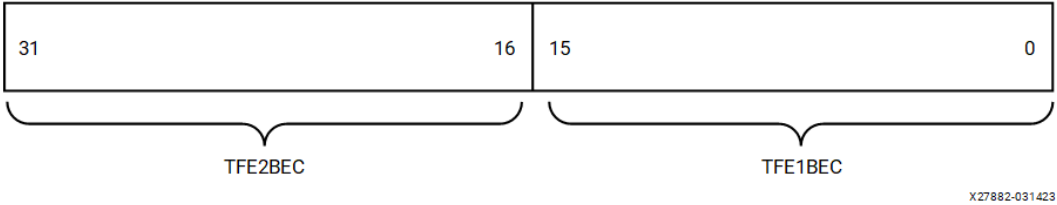
Transmit FIFO ECC Error Counter Register

The Transmit FIFO ECC Error counter register shows the transmit FIFO ECC error count. The lower 16 bits [15:0] contains the 1-bit error count and the upper 16 bits [31:16] contains 2-bit error count of the Transmit FIFO.

Table: AXI4-Stream Transit FIFO ECC Counter Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
31:16	TFE2BEC	Read	0	Transmit FIFO ECC 2-bit error counter. On reaching maximum value of 0xFFFF, it stays there until reset.
15:0	TFE1BEC	Read	0	Transmit ECC 1-bit error counter. On reaching the maximum value of 0xFFFF, it stays there until reset.

Figure: AXI4–Stream Transmit FIFO ECC Error Counter Register (Offset 0x48)



Receive FIFO ECC Configuration Register

The Receive FIFO ECC Configuration Register shown in is not an actual register, but is instead a read–write address. Bit 0 is for resetting the Receive FIFO ECC Error counters, other bits are reserved for future use.

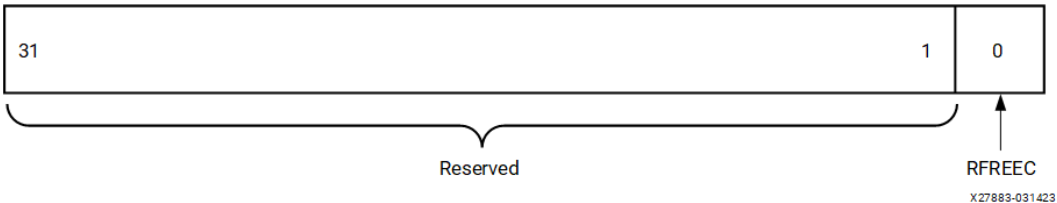
ECC Error counters Reset is applied as long as the bit 0 is '1'.

Table: AXI4–Stream Receive FIFO ECC Configuration Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
31:1	N/A	Read	0	These bits are reserved for future definition and always return all zeros.

Bit(s)	Name	Core Access	Reset Value	Description
0	RFREEC	Read/Write	0	Receive FIFO Reset ECC error counters: Reset ECC error counters for Receive FIFO. Write 1 to clear the Receive FIFO ECC Error counters. Counters are cleared as long as this bit is '1'.

Figure: AXI4–Stream Receive FIFO ECC Configuration Register (Offset 0x4C)



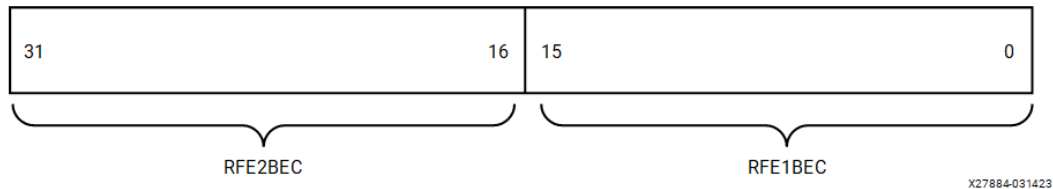
Receive FIFO ECC Error Counter Register

The Receive FIFO ECC Error counter register shows the ECC error count for Receive FIFO. The lower 16 bits [15:0] stores the 1-bit error count and the upper 16 bits [31:16] stores the 2-bit error count of the Receive FIFO.

Table: AXI4-Stream Receive FIFO ECC Error Counter Register Bit Definitions

Bit(s)	Name	Core Access	Reset Value	Description
31:16	RFE2BEC	Read	0	Receive ECC 2-bit error counter. On reaching maximum value of 0xFFFF, it stays there until reset.
15:0	RFE1BEC	Read	0	Receive ECC 1-bit error counter. On reaching maximum value of 0xFFFF, it stays there until reset.

Figure: AXI4–Stream Receive FIFO ECC Error Counter Register (Offset 0x50)



Reserved Registers

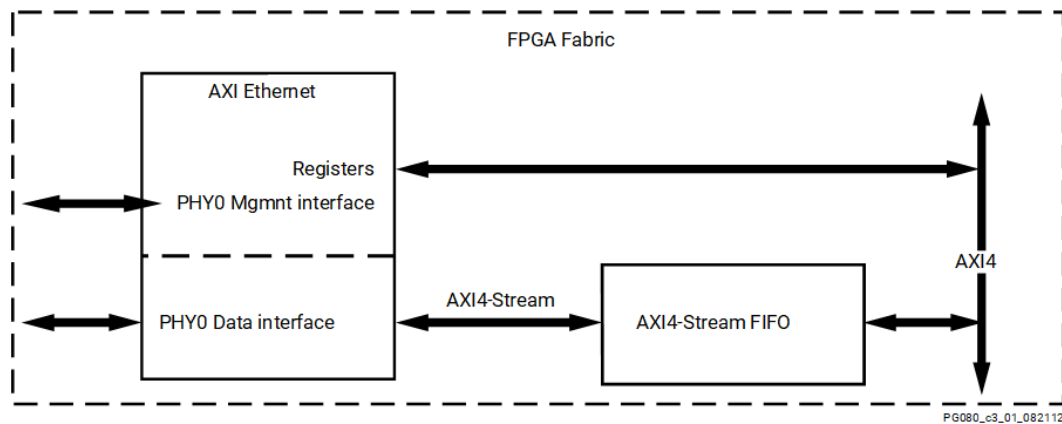
Reading from reserved registers returns zeros and writing to reserved registers have no effect. However, any accesses to address offset 0x54 and above causes undefined results.

Designing with the Core

This chapter includes guidelines and additional information to facilitate designing with the core.

General Design Guidelines

The AXI4–Stream FIFO core can be used in applications to interface between an AXI4 interface and an AXI4–Stream interface. An example of this application would be the AMD AXI Ethernet IP core which has an AXI4–Lite interface for configuration and control and an AXI4–Stream interface for data transfer. The AXI4–Stream FIFO can be used as a bridge to interface to the AXI4 or AXI4–Lite interfaces as shown in the following figure.

Figure: AXI4–Stream FIFO Connected to an AXI Ethernet Core

Design Tools

The AXI4–Stream FIFO core design is implemented using VHDL code. The AMD Vivado™ Design Suite includes a synthesis tool for synthesizing the core.

Target Technology

The target technology is an FPGA listed in the supported device family field of the IP Facts Table.

Clocking

The AXI4–Stream FIFO core operates on a single clock (`s_axi_aclk`), and all input and output interface signals of the AXI4–Stream and AXI4–Lite/AXI4 interfaces are synchronized with this clock.

Resets

The AXI4-Stream FIFO core uses a single asynchronous reset (`s_axi_aresetn`). The core stays in a reset state for three clock cycles after the reset applied.

Protocol Description

The AXI4-Stream FIFO core uses the industry standard AMBA® AXI4-Stream and AXI4 Protocol Specification. The following image details the AXI4-Stream interface where INFORMATION represents all AXI4-Stream signals except TVALID/TREADY.

Figure: AXI4-Stream Interface Timing Diagram

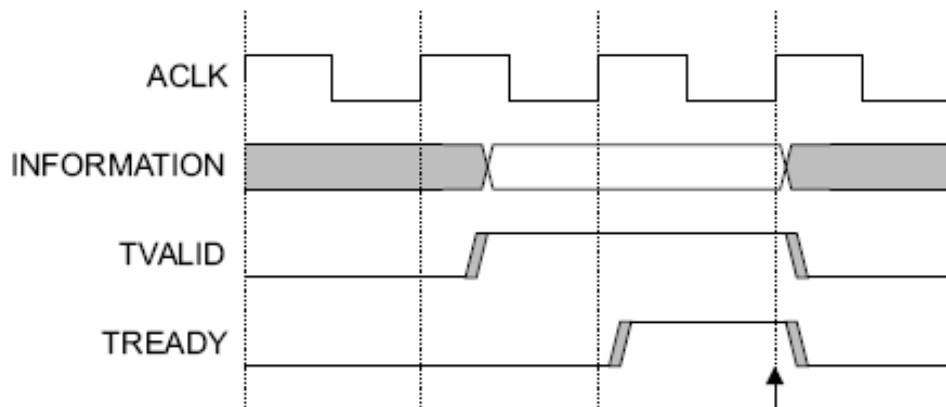
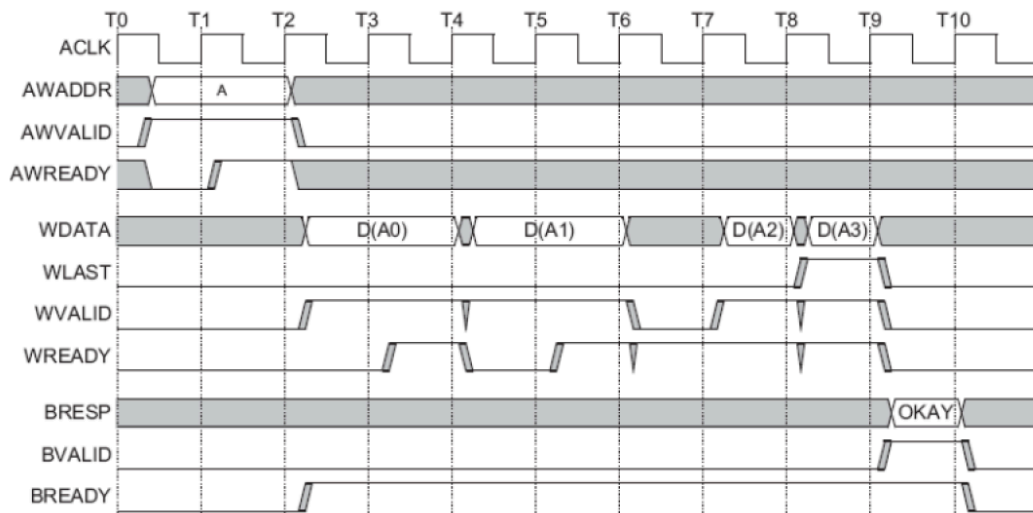
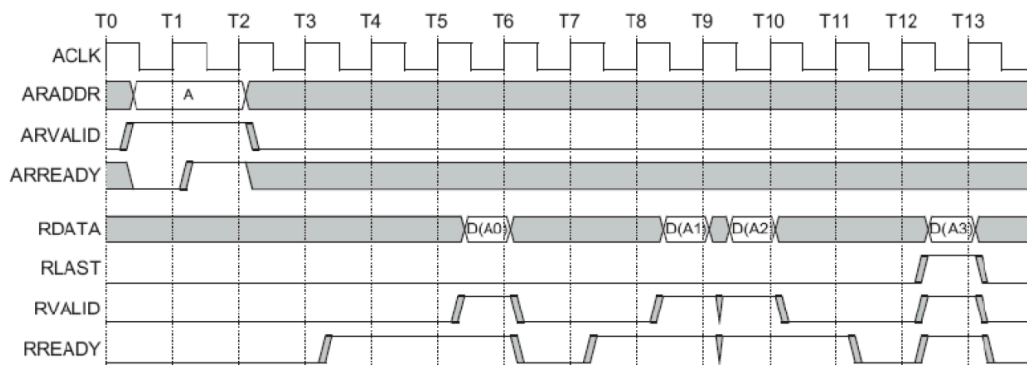


Figure: AXI4 Write Burst Transaction**Figure: AXI4 Read Burst Transaction**

This section describes the operation of the AXI4-Stream FIFO core through register accesses using the AXI Ethernet core as an example.

Depending on the Data Interface Option, either AXI4 or AXI4-Lite is used for FIFO accesses. When AXI4-Lite is selected, register access and FIFO accesses are handled by the AXI4-Lite interface. When AXI4 is selected, register access is handled by AXI4-Lite interface and FIFO accesses are

handled by AXI4 interface. Data bursting is possible when the Data Interface option is AXI4.

The AXI4-Stream FIFO supports two packet transmission modes: store-and-forward mode and cut-through mode.

- In store-and-forward mode, packet transmission begins on the AXI4-Stream interface in the following circumstances:
 - When the complete packet is written to the FIFO.
 - When the length of packet is written to TX Length Register.

In this mode, the size of the FIFO must be large enough to hold the complete packet.

In cut-through mode, packet transmission begins on the AXI4-Stream interface when there is enough data in the FIFO. In this mode, the FIFO does not need to hold the complete packet. However, ensure that the AXI4-Stream interface does not under run.

The AXI4-Stream FIFO core supports two packet receiving modes: store-and-forward mode and cut-through mode.

- In store-and-forward mode, the data from the AXI4-Stream interface is completely stored in the FIFO prior to making it available over the AXI4 interface. The RX Length Register (RLR) is updated with the length of packet, and the interrupt status registers are updated.
- In cut-through mode, packet reception begins on the AXI4 interface when there is enough data in the FIFO. In this mode, the FIFO does not need to hold the complete packet. The RX Length Register (RLR) register is updated with the data count continuously until the packet is completely received. The RLR register value can be used to read the data out of the FIFO. The cut-through mode supports reception of packets that are larger than the FIFO size. However, ensure that AXI4-Stream interface does not over-run the FIFO.

Note: The AXI4-Stream signal `tlast` along with `tvalid` denotes the end of packet in store-and-forward mode and cut-through mode.

Note: The IP is designed to hold a maximum of $(\text{RX FIFO depth} / 4) + 2$ one word packets. For example, if the RX FIFO depth is 512, then the IP

can hold a maximum of 130 one word packets.

Programing Sequence

This chapter includes details about [Programming Sequence Using Direct Register Read/Write](#) and [Programming Sequence Using Example Software Driver](#).

Programming Sequence Using Direct Register Read/Write

This programming sequence is provided for you to directly perform the read/write operations to the registers and this might not match with the example driver software that is provided along with this IP.

The following two tables illustrate a power-up read of the registers followed by a programming sequence for transmission and reception of a single packet in store-and-forward mode and cut-through mode using AXI4-Lite interface. See the register definitions for further information and options.

Table: Programming Sequence for TX and RX in Store-and-Forward Mode

Register	Access	Value	Activity
Power-up/Reset Read of Register Values			
ISR	Read Word	01D00000	Read interrupt status register (indicates transmit reset complete and receive reset complete)

Register	Access	Value	Activity
ISR	Write Word	0xFFFFFFFF	Write to clear reset done interrupt bits
ISR	Read Word	0x00000000	Read interrupt status register
IER	Read Word	0x00000000	Read interrupt enable register
TDFV	Read Word	0x000001FC	Read the transmit FIFO vacancy (for TX FIFO Depth of 512)
RDFO	Read Word	0x00000000	Read the receive FIFO occupancy
Transmit a Packet			
IER	Write Word	0x0C000000	Enable transmit complete and receive complete interrupts
TDR	Write Word	0x00000002	Transmit Destination address (0x2 = destination device address is 2)
TDFD	Write Word	0xFFFFFFFF	4 bytes of data
TDFD	Write Word	0x12345678	4 bytes of data
TDFD	Write Word	0x00010203	4 bytes of data

Register	Access	Value	Activity
TDFD	Write Word	0x08090A0B	4 bytes of data
TDFD	Write Word	0x10111213	4 bytes of data
TDFD	Write Word	0x18191A1B	4 bytes of data
TDFD	Write Word	0x20212223	4 bytes of data
TDFD	Write Word	0x28292A2B	4 bytes of data
TDFV	Read Word	0x000001F4	Read the transmit FIFO vacancy
TLR	Write Word	0x00000020	Transmit length (0x20 = 32bytes), this starts transmission
ISR	Read Word	0x08000000	A typical value after TX Complete is indicated by interrupt
ISR	Write Word	0xFFFFFFFF	Write to clear transmit complete interrupt bits
ISR	Read Word	0x00000000	Read interrupt status register
TDFV	Read Word	0x000001FC	Read the transmit FIFO vacancy
Receive a Packet			

Register	Access	Value	Activity
ISR	Read Word	0x04000000	A typical value after RX Complete is indicated by interrupt
ISR	Write Word	0xFFFFFFFF	Write to clear receive complete interrupt bits
ISR	Read Word	0x00000000	Read interrupt status register
RDFO	Read Word	0x00000008	Read the receive FIFO occupancy
RLR	Read Word	0x00000020	Receive length (0x20 = 32 bytes) indicates number of bytes to read
RDR	Read Word	0x00000002	Receive Destination address (0x2 = destination device address is 2)
RDFD	Read Word	0x00000008	Read the receive FIFO occupancy
RDFD	Read Word	0xFFFFFFFF	4 bytes of data
RDFD	Read Word	0x12345678	4 bytes of data
RDFD	Read Word	0x00010203	4 bytes of data

Register	Access	Value	Activity
RDFD	Read Word	0x08090A0B	4 bytes of data
RDFD	Read Word	0x10111213	4 bytes of data
RDFD	Read Word	0x18191A1B	4 bytes of data
RDFD	Read Word	0x20212223	4 bytes of data
RDFD	Read Word	0x28292A2B	4 bytes of packet data and CRC value
RDFO	Read Word	0x00000000	Read the receive FIFO occupancy (no further receive packets to process)

Table: Programming Sequence for TX and RX in Cut-Through Mode

Register	Access	Value	Activity
Power-up Read of Register Values			
ISR	Read Word	0x01D00000	Read interrupt status register (indicates transmit reset complete and receive reset complete)
ISR	Write Word	0xFFFFFFFF	Write to clear reset done interrupt bits
ISR	Read Word	0x00000000	Read interrupt status register

Register	Access	Value	Activity
IER	Read Word	0x00000000	Read interrupt enable register
TDFV	Read Word	0x000001FC	Read the transmit FIFO vacancy
RDFO	Read Word	0x00000000	Read the receive FIFO occupancy
Transmit a Packet in Cut-Through Mode			
IER	Write Word	0x0C000000	Enable transmit complete and receive complete interrupts
TDR	Write Word	0x00000002	Transmit Destination address (0x2 = destination device address is 2)
TDFD	Write Word	0xFFFFFFFF	4 bytes of data
TDFD	Write Word	0x12345678	4 bytes of data
TDFD	Write Word	0x00010103	4 bytes of data
TDFD	Write Word	0x08090A0B	4 bytes of data
TDFD	Write Word	0x10111213	4 bytes of data
TDFD	Write Word	0x18191A1B	4 bytes of data
TDFD	Write Word	0x20212223	4 bytes of data

Register	Access	Value	Activity
TDFD	Write Word	0x28292A2B	4 bytes of data
TLR	Write Word	0x00000020	Transmit length (0x20 = 32bytes), this starts transmission
ISR	Read Word	0x08000000	A typical value after TX Complete is indicated by interrupt
ISR	Write Word	0xFFFFFFFF	Write to clear transmit complete interrupt bits
ISR	Read Word	0x00000000	Read interrupt status register
TDFV	Read Word	0x000001FC	Read the transmit FIFO vacancy
Receive a Packet in Cut-Through Mode			
IER	Write Word	0x04100000	Enable receive complete and Receive FIFO Programmable Full (RFPF) threshold interrupts

Register	Access	Value	Activity
ISR	Read Word	0x00100000	A typical value after RFPF is indicated by interrupt
ISR	Write Word	0x00100000	Reset RFPF interrupt
RLR	Read Word	0x80000010	Read the receive FIFO occupancy. If bit-31 is 1, it indicates that a partial packet is available. 0x80000010 = 16 bytes
RDFD	Read Word	0xFFFFFFFF	Started reading partial packet. 4 bytes of data
RDFD	Read Word	0x21031987	4 bytes of data
RDFD	Read Word	0xAEF10011	4 bytes of data
RDFD	Read Word	0x27071985	4 bytes of data
ISR	Read Word	0x04000000	A typical value after RX complete is indicated by interrupt

Register	Access	Value	Activity
ISR	Write Word	0x04000000	Reset RX complete interrupt
RLR	Read Word	0x00000020	Receive length (0x20 = 32 bytes) indicates number of bytes to be read
RDR	Read Word	0x00000002	Receive Destination address (0x2 = destination device address is 2)
RDFD	Read Word	0x10101021	Reading remaining 16 bytes. 4 bytes of data
RDFD	Read Word	0x21041987	4 bytes of data
RDFD	Read Word	xAEF10011	4 bytes of data
RDFD	Read Word	0x27061985	4 bytes of data
ISR	Read Word	0x04000000	A typical value after RX complete is indicated by interrupt
ISR	Write Word	0x04000000	Reset RX complete interrupt

Register	Access	Value	Activity
RLR	Read Word	0x80000014	Read the receive FIFO occupancy. If bit 31 is 0, it indicates that a full packet is available. 0x80000014 = 20 bytes
RDR	Read Word	0x00000003	Receive Destination address (0x3 = destination device address is 3)
RDFD	Read Word	0x10101010	Reading packet. 4 bytes of data
RDFD	Read Word	0x20041981	4 bytes of data
RDFD	Read Word	xAEF1001F	4 bytes of data
RDFD	Read Word	0x21021958	4 bytes of data
RDFD	Read Word	0x0E0CB231	4 bytes of data
RDFO	Read Word	0x00000000	Read the receive FIFO occupancy

Programming Sequence Using Example Software Driver

Init Sequence

Calls the function `XL1Fifo_CfgInitialize` for the initialization that does a reset of the TX/RX registers and then clear the ISR.

Transmit a Packet

1. Call `SetupInterruptSystem` function which does the initialization of the interrupt controller
2. Call the function `XL1Fifo_IntEnable` to enable all the required interrupts.
3. Call the `TxSend` function to write the data to TXFIFO and check TDFV for the FIFO occupancy before writing to the TX FIFO using the `XL1Fifo_iTxVacancy` function.
4. Start transmission by writing to TLR using the `XL1Fifo_iTxSetLen` function.
5. Wait for the data transmission to complete. Calls `FIFOHandler` interrupt handler and if it is a TX complete, calls `FIFOSendHandler`; next, clears the ISR.

Receive a Packet

1. On receiving the RX Interrupt, the `FIFOHandler` interrupt handler is called which in turn calls the `FIFORecvHandler`.
2. Read RLR to find the number of bytes.
3. Read the number of bytes till the number of bytes read from RLR while checking the occupancy by using the `XL1Fifo_iRxOccupancy` function.
4. Clear the ISR.
5. Check if there is an Interrupt pending for RX. Go to step 2.

Design Flow Steps

This section describes customizing and generating the core, constraining the core, and the simulation, synthesis, and implementation steps that are

specific to this IP core. More detailed information about the standard AMD Vivado™ design flows and the IP integrator can be found in the following Vivado Design Suite user guides:

- Vivado Design Suite User Guide: Designing IP Subsystems using IP Integrator ([UG994](#))
- Vivado Design Suite User Guide: Designing with IP ([UG896](#))
- Vivado Design Suite User Guide: Getting Started ([UG910](#))
- Vivado Design Suite User Guide: Logic Simulation ([UG900](#))

Customizing and Generating the Core

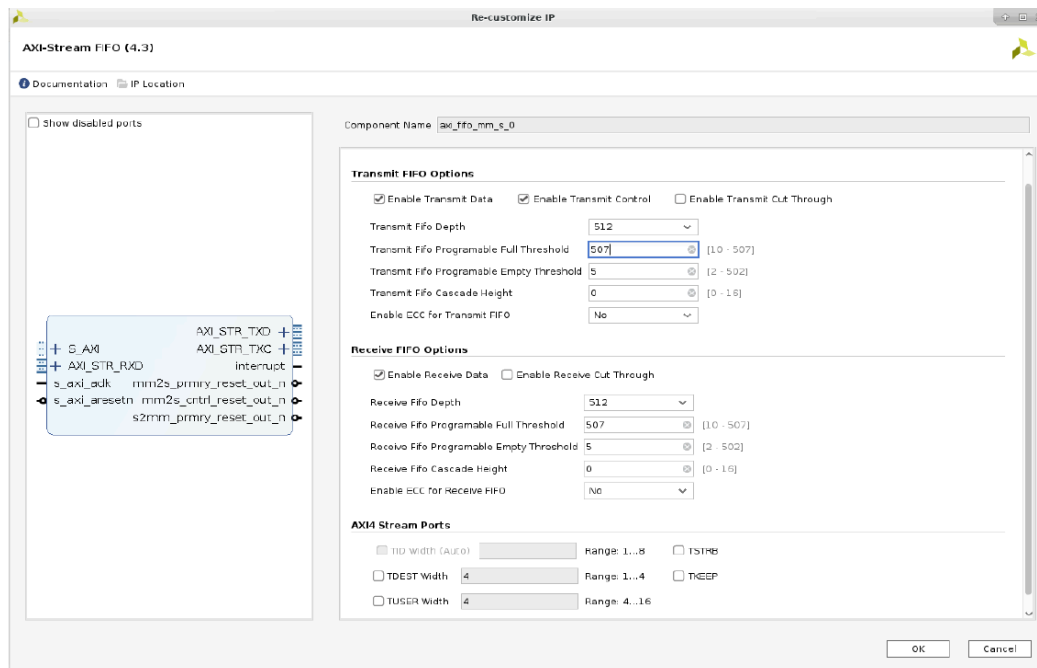
This section includes information about using AMD Adaptive Computing tools to customize and generate the core in the AMD Vivado™ Design Suite. You can customize the IP for use in your design by specifying values for the various parameters associated with the IP core using the following steps:

1. Select the IP from the IP catalog. The AXI4–Stream FIFO core is located under AXI Infrastructure in the AMD Vivado™ IP catalog.
2. Double-click the selected IP or select the Customize IP command from the toolbar or right-click menu.

For details, see the Vivado Design Suite User Guide: Designing with IP ([UG896](#)) and the Vivado Design Suite User Guide: Getting Started ([UG910](#)).

Note: Figures in this chapter are illustrations of the AMD Vivado™ Integrated Design Environment (IDE). This layout might vary from the current version.

Figure: Vivado IDE for AXI4–Stream FIFO



The above screenshot illustrates the Re-customize IP screen for AXI4–Stream FIFO.

- **Data Interface**

AXI4–Lite interface is for register access and transmit/receive FIFO accesses. Supported data width in this mode is 32. In this mode, all register accesses, except transmit/receive data FIFO registers are accessed using the AXI4–Lite interface. Only transmit/receive data FIFO registers are accessed using the AXI4 interface. Data bursting is possible only in the AXI4 mode. The supported data width in this mode is 32, 64, 128, 256, and 512 (only for transmit/receive data FIFO registers).

- **AXI4 Data Width**

The AXI4 data width defines the width of the transmit/receive data FIFO registers. Supported data width is 32, 64, 128, 256, and 512.

- **AXI4 ID Width**

The AXI4 ID width defines the width of the AWID/BID/ARID/RID ports. Supported ID width is 0 to 32.

- **Transmit FIFO**

Enable Transmit Data: Enables the transmit datapath (from AXI4 interface to AXI4–Stream interface).

Enable Transmit Control: Enables the transmit control. The AXI4–Stream Transmit Control Interface supports the transmit protocol of AXI Ethernet cores.

Enable Transmit Cut–Through: Enables the cut–through mode in which packet transmission begins on the AXI4–Stream interface when there is enough data in the FIFO. When this option is not selected, the FIFO operates in Store–and–Forward Mode.

Transmit FIFO Depth: Valid range of the transmit FIFO depth is 512 to 128 k locations (powers of 2).

Transmit FIFO Programmable Full Threshold: The valid range for this threshold is provided in the IDE. When the difference between the read and write pointers of the transmit FIFO reaches the programmable FULL threshold value, the TFPF bit in ISR is set.

Transmit FIFO Programmable Empty Threshold: The valid range for this threshold is provided in the IDE. When the difference between the read and write pointers of the transmit FIFO reaches the programmable EMPTY threshold value, the TFPE bit in ISR is set.

Transmit FIFO Cascade Height: This parameter specifies the number of block RAMs present in one cascade chain of a Transmit FIFO. To implement a memory that requires more than 1 block RAM, it needs to connect several block RAMs using built–in cascade and/or fabric LUTs (MUXes). When `cascade_height = 1`, synthesis uses NO cascade at all. This is used for maximum timing performance. Default value is 0 (that is, synthesis has an option to choose cascading or not).

Enable ECC for Transmit FIFO: This parameter enables the ECC on Transmit FIFO. If any 1–bit error is detected on Transmit FIFO, TFE1BE (Transmit FIFO ECC 1 Bit Error) bit in the ISR is set. 1–bit error is corrected automatically. Number of 1–bit errors are counted and stored in Transmit FIFO ECC Error Counter register lower 16 bits (for example [15:0]). If any 2–bit error is detected on Transmit FIFO, TFE2BE (Transmit

FIFO ECC 2 Bit Error) bit in the ISR is Set and the number of 2-bit errors are counted and stored in Transmit FIFO ECC Error Counter register upper 16 bits (for example [31:16]). The default value is 0.

- **Receive FIFO**

 **Note:** The Full and Empty threshold range is different from previous version of the core (v4.1) due to the use of XPM libraries. See the IP catalog GUI for the valid range of values.

Enable Receive Data Enables the receive datapath (from AXI4-Stream interface to AXI4 interface).

Enable Receive Cut-Through: Enable Receive Cut-Through: Enables the cut-through mode in which packet reception begins on the AXI4 interface when there is enough data in the FIFO. When this option is not selected, the FIFO operates in Store-and-Forward Mode.

Receive FIFO Depth: Valid range of the receive FIFO depth is 512 to 128 k locations (powers of 2).

Receive FIFO Programmable Full Threshold: The valid range for this threshold is provided in the IDE. When the difference between the read and write pointers of the receive FIFO reaches the programmable FULL threshold value, the RFPF bit in ISR is set.

Receive FIFO Programmable Empty Threshold: The valid range for this threshold is provided in the IDE. When the difference between the read and write pointers of the receive FIFO reaches the programmable EMPTY threshold value, the RFPE bit in ISR is set.

Receive FIFO Cascade Height: This parameter specifies the number of block RAMs present in one cascade chain of a Receive FIFO. To implement a memory that requires more than 1 block RAM, it needs to connect several block RAMs using built-in cascade and/or fabric LUTs (MUXes). When `cascade_height = 1`, synthesis uses NO cascade at all. This is used for maximum timing performance. Default value is 0 (that is, synthesis has an option to choose cascading or not).

Enable ECC for Receive FIFO: This parameter enables the ECC on Receive FIFO. If any 1-bit error is detected on Receive FIFO, RFE1BE (Receive FIFO ECC 1 Bit Error) bit in the ISR is set. 1-bit error is corrected automatically. Number of 1-bit errors are counted and stored in the Receive FIFO ECC Error Counter register lower 16 bits (for example [15:0]). If any 2-bit error is detected on Receive FIFO, RFE2BE (Receive FIFO ECC 2-Bit Error) bit in the ISR is set and the number of 2-bit errors

are counted and stored in Receive FIFO ECC Error Counter register upper 16 bits, for example [31:16]. Default value is 0.

AXI4–Stream Ports: The AXI4–Stream FIFO configures the widths for TUSER, TID and TDEST signals. The valid range of these signals is provided in the IDE. For TKEEP and TSTRB signals, the width is determined by the configured DATA width and is internally calculated by using the equation (DATA Width)/8.

User Parameters

The following table shows the relationship between the GUI fields in the Vivado IDE and the User Parameters (which can be viewed in the Tcl Console).

Table: GUI Parameter to User Parameter Relationship

GUI Parameter/Value	User Parameter/Value	Default Value
Component Name	Component_Name	None
AXI4 ID Width	C_S_AXI_ID_WIDTH	4
Data Interface	C_DATA_INTERFACE_TYPE	AXI4–Lite
AXI4 Data Width	C_S_AXI4_DATA_WIDTH	32
Transmit FIFO Options		
Enable Transmit Data	C_USE_TX_DATA	True: 1
Enable Transmit Control	C_USE_TX_CTRL	True: 1
Enable Transmit Cut Through	C_USE_TX_CUT_THROUGH	False: 0
Transmit FIFO Depth	C_TX_FIFO_DEPTH	512

GUI Parameter/Value	User Parameter/Value	Default Value
Transit FIFO Programmable Full Threshold	C_TX_FIFO_PF_THRES HOLD	507
Transmit FIFO Programmable Empty Threshold	C_TX_FIFO_PE_THRES HOLD	5
Transmit FIFO Cascade Height	C_TX_CASCADE_HEIGHT	0
Enable ECC for Transmit FIFO	TX_ENABLE_ECC	0
Receive FIFO Options		
Enable Receive Data	C_USE_RX_DATA	True: 1
Enable Receive Cut Through	C_USE_RX_CUT_THROUGH	False: 0
Receive FIFO Depth	C_RX_FIFO_DEPTH	512
Receive FIFO Programmable Full Threshold	C_RX_FIFO_PF_THRES HOLD	507
Receive FIFO Programmable Empty Threshold	C_RX_FIFO_PE_THRES HOLD	5
Receive FIFO Cascade Height	C_RX_CASCADE_HEIGHT	0
Enable ECC for Receive FIFO	RX_ENABLE_ECC	0
AXI4-Stream Ports		
	C_HAS_AXIS_TUSER	False: 0

GUI Parameter/Value	User Parameter/Value	Default Value
	C_HAS_AXIS_TID	False: 0
	C_HAS_AXIS_TDEST	False: 0
TID Width	C_AXIS_TID_WIDTH	4
TUSER Width	C_AXIS_TUSER_WIDTH	4
TDEST Width	C_AXIS_TDEST_WIDTH	4
TSTRB	C_HAS_AXIS_TSTRB	False: 0
TKEEP	C_HAS_AXIS_TKEEP	False: 0

Output Generation

For details, see the Vivado Design Suite User Guide: Designing with IP ([UG896](#)).

Constraining the Core

There are no constraints associated with this core.

Simulation

For comprehensive information about AMD Vivado™ simulation components, as well as information about using supported third party tools, see the Vivado Design Suite User Guide: Logic Simulation ([UG900](#)).

 **Important:** The Full and Empty threshold range is different from previous version of the core (v4.1) due to the use of XPM libraries. See the IP catalog for the valid range of values.

Synthesis and Implementation

For details about synthesis and implementation, see the Vivado Design Suite User Guide: Designing with IP ([UG896](#)).

Verification, Compliance, and Interoperability

This chapter provides details about how this IP core was tested for compliance.

Simulation

The AXI4-Stream FIFO core has been tested with AMD Vivado™ simulator, and Mentor Graphics QuestaSim simulator.

Hardware Testing

The AXI4-Stream FIFO core has been hardware validated at 200 MHz on a KC705 board using AMD Kintex™ 7 –2 speed grade device (325T). The IP was configured for AXI4-Lite to work with AMD AXI Ethernet IP.

Upgrading

This appendix contains information about migrating a design from ISE to the AMD Vivado™ Design Suite, and for upgrading to a more recent version of the IP core. For customers upgrading in the AMD Vivado™ Design Suite, important details (where applicable) about any port changes and other impact to user logic are included.

Migrating to the Vivado Design Suite

For information about migrating to the AMD Vivado™ Design Suite, see the ISE to Vivado Design Suite Migration Guide ([UG911](#)).

Upgrading in the Vivado Design Suite

This section provides information about any changes to the user logic or port designations that take place when you upgrade to a more current version of this IP core in the AMD Vivado™ Design Suite.

Parameter Changes

There are two new parameters added: Enable ECC for Transmit FIFO and Enable ECC for Receive FIFO.

Port Changes

There are no port changes from the previous release to this release.

Debugging

This appendix includes details about resources available on the AMD Support website and debugging tools.

If the IP requires a license key, the key must be verified. The AMD Vivado™ design tools have several license checkpoints for gating licensed IP through the flow. If the license check succeeds, the IP can continue generation. Otherwise, generation halts with an error. License checkpoints are enforced by the following tools:

- Vivado Synthesis
- Vivado Implementation
- write_bitstream (Tcl command)

 **Note:** IP license level is ignored at checkpoints. The test confirms a valid license exists. It does not check IP license level.

Finding Help with AMD Adaptive Computing Solutions

To help in the design and debug process when using the core, the [Support web page](#) contains key resources such as product documentation, release notes, answer records, information about known issues, and links for obtaining further product support. The [Community Forums](#) are also available where members can learn, participate, share, and ask questions about AMD Adaptive Computing solutions.

Documentation

This product guide is the main document associated with the core. This guide, along with documentation related to all products that aid in the

design process, can be found on the [AMD Adaptive Support web page](#) or by using the AMD Adaptive Computing Documentation Navigator. Download the Documentation Navigator from the [Downloads page](#). For more information about this tool and the features available, open the online help after installation.

Answer Records

Answer Records include information about commonly encountered problems, helpful information on how to resolve these problems, and any known issues with an AMD Adaptive Computing product. Answer Records are created and maintained daily to ensure that users have access to the most accurate information available.

Answer Records for this core can be located by using the Search Support box on the main [AMD Adaptive Support web page](#). To maximize your search results, use keywords such as:

- Product name
- Tool message(s)
- Summary of the issue encountered

A filter search is available after results are returned to further target the results.

Master Answer Record for the Core

AR [54447](#).

Technical Support

AMD Adaptive Computing provides technical support on the [Community Forums](#) for this AMD LogiCORE™ IP product when used as described in the product documentation. AMD Adaptive Computing cannot guarantee timing, functionality, or support if you do any of the following:

- Implement the solution in devices that are not defined in the documentation.
- Customize the solution beyond that allowed in the product documentation.
- Change any section of the design labeled DO NOT MODIFY.

To ask questions, navigate to the [Community Forums](#).

Debug Tools

There are many tools available to address AXI4–Stream FIFO core design issues. It is important to know which tools are useful for debugging various situations.

Vivado Design Suite Debug Feature

The AMD Vivado™ Design Suite debug feature inserts logic analyzer and virtual I/O cores directly into your design. The debug feature also allows you to set trigger conditions to capture application and integrated block port signals in hardware. Captured signals can then be analyzed. This feature in the Vivado IDE is used for logic debugging and validation of a design running in AMD devices.

The Vivado logic analyzer is used to interact with the logic debug LogiCORE IP cores, including:

- ILA 2.0 (and later versions)
- VIO 2.0 (and later versions)

See the Vivado Design Suite User Guide: Programming and Debugging ([UG908](#)).

Additional Resources and Legal Notices

Finding Additional Documentation

Technical Information Portal

The AMD Technical Information Portal is an online tool that provides robust search and navigation for documentation using your web browser. To access the Technical Information Portal, go to <https://docs.amd.com>.

Documentation Navigator

Documentation Navigator (DocNav) is an installed tool that provides access to AMD Adaptive Computing documents, videos, and support resources, which you can filter and search to find information. To open DocNav, do the following:

- From the AMD Vivado™ IDE, select Help > Documentation and Tutorials.
- On Windows, click the Start button and select AMDDesignTools > DocNav.
- At the Linux command prompt, enter `docnav`.

 **Note:** For more information on DocNav, refer to the Documentation Navigator User Guide ([UG968](#)).

Design Hubs

AMD Design Hubs provide links to documentation organized by design tasks and other topics, which you can use to learn key concepts and address frequently asked questions. To access the Design Hubs, do the following:

- In DocNav, click the Design Hubs View tab.
- Go to the [Design Hubs](#) web page.

Support Resources

For support resources such as Answers, Documentation, Downloads, and Forums, see [Support](#).

References

These documents provide supplemental material useful with this guide:

1. [AMBA AXI4–Stream Protocol Specification](#)
2. [AMBA AXI4 Protocol Specification](#)
3. Vivado Design Suite User Guide: Designing IP Subsystems using IP Integrator ([UG994](#))
4. AXI 1G/2.5G Ethernet Subsystem Product Guide ([PG138](#))
5. [LogiCORE IP AXI Slave Burst \(DS769\)](#)
6. Virtex–6 Family Overview ([DS150](#))
7. 7 Series FPGAs Data Sheet: Overview ([DS180](#))
8. Vivado Design Suite User Guide: Designing with IP ([UG896](#))
9. Vivado Design Suite User Guide: Getting Started ([UG910](#))
10. Vivado Design Suite User Guide: Logic Simulation ([UG900](#))
11. Vivado Design Suite User Guide: Implementation ([UG904](#))
12. ISE to Vivado Design Suite Migration Guide ([UG911](#))
13. Vivado Design Suite User Guide: Programming and Debugging ([UG908](#))

Revision History

The following table shows the revision history for this document.

Section	Revision Summary
01/09/2026 Version 4.3	
Transmit Data FIFO Vacancy Register (TDFV)	Added a note.
11/08/2023 Version 4.3	
Multiple Sections	Fixed discrepancies between PDF and HTML documents.

Section	Revision Summary
05/17/2023 Version 4.3	
Features	Added ECC enable/disable option for TX and RX FIFO.
Product Specification	Added the following new registers: Transmit FIFO, ECC Configuration register, Transmit FIFO ECC Error Counter register, Receive FIFO ECC Configuration register and Receive FIFO ECC Error Counter register.
Customizing and Generating the Core	Added Enable ECC for Transmit, FIFO parameter and Enable ECC for Receive FIFO parameter.
Customizing and Generating the Core	Updated section.
10/30/2019 Version 4.2	
IP Facts	Added families to Supported Device Family and added Release Notes and Known Issues.
Customizing and Generating the Core	Added Transmit/Receive FIFO Cascade Height descriptions.
Table 1	Added Transmit/Receive FIFO Cascade Height.
05/22/2019 Version 4.2	
Features	Updated configurable data width.
Protocol Description	Added RX FIFO note.
Transmit Data FIFO Reset Register (TDFR)	Added description on inactive reset.

Section	Revision Summary
Receive Data FIFO Reset Register (RDFR)	Added description on inactive reset.
Customizing and Generating the Core	Updated supported data width for AXI4 and AXI4 Data Width.
12/05/2018 Version 4.2	
Features	Updated standalone driver links.
Customizing and Generating the Core	Updated section.
Table 1	Updated TX and RX FIFO Programmable Empty Threshold default value to 5.
04/06/2016 Version 4.1	
Interrupt Status Register Bit Definitions	Corrected section, including interrupt signal definitions and configurable data widths.
04/01/2015 Version 4.1	
Table 2	Updated the read interrupt status register (ISR) value.
Table 1	Updated the Reset Values for Receive FIFO Programmable Full (RFPF) and Transmit FIFO Programmable Full (TFPF). Updated the Transmit Length Register (TLR) width to support packets up to 8 Mbytes-4. Updated the Transmit Length Register (TLR) to support packets up to 8388604 in length.

Section	Revision Summary
10/01/2014 Version 4.1	
Register Space	Updated the following register definitions: • Transmit Data FIFO Vacancy Register (TDFV) • Transmit Data FIFO Data Write Port (TDFD) • Receive Data FIFO Occupancy Register (RDFO) • Receive Data FIFO Data Read Port (RDFD) • Transmit Length Register (Store-and-Forward Mode) • Transmit Length Register (Cut-Through Mode) • Receive Length Register (Store-and-Forward Mode) • Receive Length Register (Cut-Through Mode) • Updated AXI4 ID Width to 0–16. • Updated the Transmit and Receive FIFO Depth ranges to 512 – 128 k (powers of 2). • Offset address for TDFD and RDFD registers changed from 0x10 to 0x0000 and 0x20 to 0x1000.
Port Descriptions	Added data widths of 128, 256, and 512 for AXI4 Data Interface.
04/02/2014 Version 4.0	

Section	Revision Summary
Register Space	Corrected the value for the Interrupt Status Register (ISR). Changed the name of the Receive Length FIFO (RLF) register to Receive Length register (RLR). Updated the maximum packet length of Transmit Data FIFO Data Write Port and the Receive Data FIFO Data Read Port. Added details about Store-and-Forward and Cut-Through modes to the Receive Length and Transmit Length registers.
12/18/2013 Version 4.0	
Multiple Sections	Added support for AMD UltraScale architecture.
10/02/2013 Version 4.0	
Multiple Sections	Added details to AMD Vivado™ Integrated Design Environment (IDE). Added selectable transmit and receive path. Added support for IP integrator.
03/20/2013 Version 3.0	
Multiple Sections	Updated core to v4.0. Removed support for ISE Design Suite.

Section	Revision Summary
Debugging	Updated section
12/18/2012 Version 2.0	
Multiple Sections	Updated core to v3.00b and Vivado Design Suite for 2012.4.
Transmit Length Register (TLR)	Updated the lengths.
Receive Length Register (RLR)	Updated the lengths.
Table 1	Clarified the ports that are not used by the core.
10/16/2012 Version 1.0	
Multiple Sections	Initial release as a product guide. Replaces DS806, LogiCORE IP AXI4–Stream FIFO Data Sheet. Changed the size of the FIFO to 508 words. Added support for an AXI4 interface, TX cut-through mode, and a 64–bit data path.

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